Environmental drivers of *Plasmodium vivax* malaria in Manaus: Modeling the impact of climate and forest loss

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ABSTRACT

Malaria persists in the Brazilian Amazon despite decades of control, and how climate and land use act on it remains hard to separate from the vector and human dynamics they mediate. We analyze *Plasmodium vivax* transmission in rural Manaus (2017-2023) and show that the delayed SIR-SEI model of Parham and Michael (2010) cannot reproduce the observed persistence, since recruitment cannot offset mosquito mortality. We reformulate it as a climate-driven SEIRS-SEI model with density-dependent recruitment, humidity-dependent adult survival and Brière-type sporogonic development, calibrated year by year, which reproduces the observed inter- and intra-annual variability. The reproduction number inherits the environmental state, so invasion risk varies within the year, and the local sensitivity attributes it to mosquito survival and the biting rate, not to the human incubation period or immunity waning. Releasing the constant per-bite probabilities to have weekly values reduced the error in every year, indicating that constant coefficients cannot reproduce the seasonal structure. A random forest on temperature, rainfall, humidity, deforestation and fire recovers it within years, but not between them. Environmental drivers therefore act mainly by modulating time-varying per-bite transmission probabilities, and because the parameters that decide threshold crossing differ from those that set the wave height, interventions ranked from the reproduction number alone may be misleading.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria in the Amazon [2]. The removal of forest canopy alters local microclimates, increasing sunlight exposure and ambient temperature, which favor the development of mosquito larvae and pupae. The primary Amazonian vector, *Anopheles darlingi*, thrives in partially deforested environments, where sunlit water bodies promote larval growth and the availability of food sources such as algae [3, 4]. Forest fragmentation accelerates mosquito development cycles and increases vectorial capacity and survivorship through higher temperature and light availability [4, 5], while forest fires, often used for land clearing, modify habitat structure and may promote subsequent fires through understory drying and fuel accumulation, contributing to long-term ecological changes that influence mosquito populations and disease transmission dynamics [6, 7, 8].

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These processes also shape human exposure to malaria. Activities such as logging, agriculture, and infrastructure development concentrate human populations in recently altered environments, often in close proximity to mosquito breeding sites, raising human-vector contact rates and facilitating transmission, particularly among non-immune migrants [9, 10]. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation. Climatic variables such as temperature and rainfall, together with human disturbances such as mining, further influence vector abundance, pathogen development, and transmission intensity. Despite the recognized importance of these environmental drivers, integrating them into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human-vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we analyze and develop modifications to a SIR-SEI model for malaria transmission in the Amazon, inspired by the host-vector framework of Parham and Michael (2010) [11] and the Trajetorias Project [12], which is a dataset of environmental epidemiological and epidemiological indicators, among others, for all municipalities of the Brazilian Legal Amazon. Our analysis focuses on the municipality of Manaus, the largest urban centre of the Brazilian Amazon and that concentrates the highest absolute number of malaria cases in the region. Within the municipality, however, the burden is concentrated in the rural zone, where incidence is substantially higher than in the urban zone or in the

municipality as a whole [12], and therefore it was decided to use the rural population of Manaus for the development of our model. The original formulation as proposed by Parham and Michael was not capable of explaining the transmission dynamics we observed in Manaus, and with the modifications described below, our model sets out to do so. The observed series of active infectious cases in Manaus over the 2017-2023 period displays pronounced inter- and intra-annual variability, with recurrent seasonal waves whose timing and amplitude change from year to year, which motivates the representation of all of the variability of the environmental drivers and, ultimately, the modifications to the baseline model described below. The model incorporates environmental variables such as temperature, rainfall, humidity, deforestation, and fire activity. Climatic factors, including temperature T, precipitation R, and humidity H, are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Deforestation and fire activity are not embedded as mechanistic drivers within the epidemiological model; instead, they are treated as environmental variables that describe the context in which transmission occurs and are used, together with T, R, and H, as predictors in the regression of the weekly per-bite transmission probabilities b_1 and b_2 (see Section 2.10). Land-use and fire activity thereby modulate transmission indirectly, through the empirical relation between environmental conditions and the effective transmission coefficients, rather than by directly altering the mechanistic mosquito rates.

Deforestation was obtained from the DETER system, whose daily alerts were aggregated at weekly resolution, with the 4-week total degraded area (clear cut, forest degradation, and burn scar combined) used as the covariate D_f [13]. Fire activity was represented by two complementary sources: the weekly number of fire hotspots detected by INPE, used as the covariate F_c [14], and annual burned-area trends from the MapBiomas project, assigned as a monthly covariate F_t [15]. Daily temperature, precipitation, and humidity were obtained from the Mosqlimate API [16] and included as the covariates T, R, and H. Together, the variables T, R, H, D_f, F_c, F_t form the environmental predictor set relating observed environmental conditions to the weekly transmission probabilities b_1 and b_2.

This paper is organized as follows: in Section 2 we present the data and the evolution of the SEIRS-SEI model, starting from the baseline delayed SIR-SEI model all the way to the proposed SEIRS-SEI construction; in Section 3 we present the mathematical analysis of the model, regarding the mosquito population dynamics, the basic reproduction number and the sensitivity analysis; in Section 4 we present the found results; which are discussed in Section 5, and we conclude in Section 6.

2. Methodology

2.1. Epidemiological Data

This study focuses on malaria cases caused by *Plasmodium vivax*, which accounts for the majority of malaria infections in Brazil and is the predominant species in the Amazon region [17, 18, 19]. Epidemiological data were obtained from the SIVEP-Malaria system [20], which provides daily records of notified malaria cases for all Brazilian municipalities from 2003 to 2023. The analysis was restricted to the period 2017-2023, for which complete epidemiological and environmental data were available. Figure 1 presents the total number of active infectious cases observed in rural Manaus over this period. This zone of the municipality was chosen over the urban or total population due to its higher incidence of cases during the periods studied by the Trajetorias Project [12]. The pronounced inter- and intra-annual variability of this series constitutes the central modelling challenge addressed in this work and motivates the modifications to the baseline model described below.

2.2. Population Estimates

As noted before, the municipality of Manaus was selected due to its high incidence of *P. vivax* malaria, particularly in rural areas. However, detailed yearly estimates of rural population were not directly available from census data. To address this limitation, population estimates were derived using data from the Trajetorias Project [12], which provides incidence values aggregated over five-year periods.

The incidence is defined as

$$\mathrm{Inc}(d, m, z, t_1, t_2) = \frac{\mathrm{Cases}(d, m, z, t_1, t_2)}{\mathrm{Pop}(m, z, (t_1 + t_2)/2) \cdot (t_2 - t_1)} \times 10^5, \tag{1}$$

where $\mathrm{Cases}(d, m, z, t_1, t_2)$ denotes the number of reported cases for disease d in municipality m and zone z over the interval $[t_1, t_2]$, and $\mathrm{Pop}(m, z, \cdot)$ represents the population at the midpoint of the interval.

This relationship was inverted in order to estimate a ratio of rural to total population from reported incidence and case counts in both zones. Using this approach, the rural population of Manaus was estimated to represent less than 1% of the total population, ranging between 0.3% and 0.6% of the total population of the municipality, while accounting for a disproportionately large share of malaria cases (approximately 30% to 42% over the study periods).

Yearly total population values for Manaus were obtained via interpolation based on official census data from the Brazilian Institute of Geography and Statistics (IBGE) for the years 2000, 2010, and 2022 [21, 22, 23]. Intermediate yearly values were constructed using linear interpolation, while values beyond 2022 were extrapolated based on observed trends.

It must be noted that the inter-census estimates of the rural

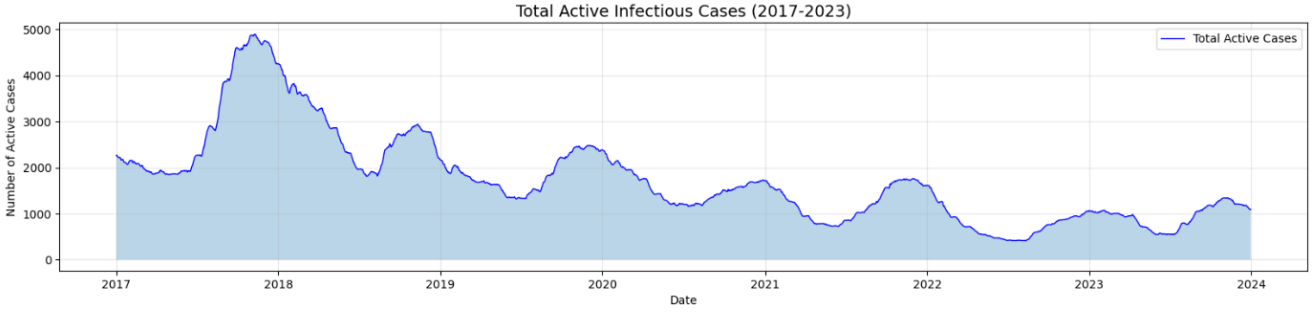


Figure 1: Total number of active infectious cases of malaria (*P. vivax*, SIVEP-Malaria) in rural Manaus, 2017-2023. The strong inter- and intra-annual variability of the disease burden is the central modelling challenge addressed in this work, and motivates the representation of all of the variability of the environmental drivers as well as the modifications to the baseline model.

population were derived from fixed rural-urban proportions based on the 2010 census. [12]. While this approach ensures internal consistency, it does not account for the marked demographic changes in Manaus over time, particularly the rapid urbanization observed in the region. To mitigate this limitation, the rural proportions used in this study were recalculated by interpolating and extrapolating the available census data, giving an approximate decrease of 2.38% of the rural population per year, providing a more realistic representation of the temporal evolution of the population. This adjustment is relevant because rural areas contribute disproportionately to malaria incidence. Nevertheless, the resulting decrease in N_y during 2017-2023 reflects a demographic trend that may still not correspond to the actual exposure of the human population to mosquitoes.

2.3. Baseline Delayed SIR-SEI Model

We begin the construction of the model with the classical host-vector framework for malaria transmission, based on the model introduced by Parham and Michael (2010) [11], which incorporates temperature- and rainfall-dependent mosquito dynamics. The human population is divided into susceptible, infected, and recovered compartments, denoted by S_H, I_H, and R_H, respectively. The mosquito population is structured into susceptible, exposed, and infectious compartments, denoted by S_M, E_M, and I_M.

The transmission dynamics are governed by the system of delay differential equations:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M(t)}{N} S_H(t) \tag{2a}$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M(t-\tau_H)}{N} S_H(t-\tau_H) - \gamma I_H(t) \tag{2b}$$

$$\frac{dR_H}{dt} = \gamma I_H(t) \tag{2c}$$

$$\frac{dS_M}{dt} = b(R,T) - \left(ab_1 \frac{I_H(t)}{N} + \mu(T) \right) S_M(t) \tag{2d}$$

$$\begin{aligned} \frac{dE_M}{dt} &= ab_1 \frac{I_H(t)}{N} S_M(t) - \mu(T) E_M(t) \\ &\quad - ab_1 \frac{I_H(t-\tau_M)}{N} S_M(t-\tau_M) \ell(\tau_M) \end{aligned} \tag{2e}$$

$$\begin{aligned} \frac{dI_M}{dt} &= ab_1 \frac{I_H(t-\tau_M)}{N} S_M(t-\tau_M) \ell(\tau_M) \\ &\quad - \mu(T) I_M(t) \end{aligned} \tag{2f}$$

All parameters are defined in Table A.1, which gives their functional forms and definitions, while state variables are defined in Table A.2, which gives their definitions, units and sources.

Although this formulation incorporates biologically relevant delays, numerical simulations revealed that, under the parameterization considered, the model fails to reproduce sustained transmission dynamics. In particular, both the infected human population and the mosquito population tends to extinction, indicating that the structure does not adequately capture endemic persistence.

2.4. ODE approximation and structural limitations

To simplify the analysis and avoid the technical complexity of delay differential equations, an ordinary differential equation (ODE) approximation was initially considered. In this formulation, the extrinsic incubation period in mosquitoes is modeled via a transition rate b_3, replacing the explicit delay τ_M.

The resulting system is given by:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M}{N} S_H \tag{3a}$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M}{N} S_H - \gamma I_H \tag{3b}$$

$$\frac{dR_H}{dt} = \gamma I_H \tag{3c}$$

$$\frac{dS_M}{dt} = b(R,T) - \left(ab_1 \frac{I_H}{N} + \mu \right) S_M \tag{3d}$$

$$\frac{dE_M}{dt} = ab_1 \frac{I_H}{N} S_M - (\mu + b_3 \ell) E_M \tag{3e}$$

$$\frac{dI_M}{dt} = b_3 \ell E_M - \mu I_M \tag{3f}$$

While this formulation restores positivity and simplifies

numerical implementation, it exhibits a fundamental limitation: the system tends toward a disease-free equilibrium after a single outbreak, failing to capture the recurrent nature of malaria transmission observed in endemic regions. This behavior highlights the importance of explicitly modeling incubation periods and immunity dynamics.

2.5. SEIRS-SEI model with ODE approximation

To address the limitations of the previous formulations, we introduce an ODE based SEIRS-SEI model that incorporates both incubation delays as done above and temporary immunity in the human population. The human compartments are extended to include an exposed class E_H, and recovered individuals lose immunity at rate ω, returning to the susceptible class.

To account for incubation dynamics in both hosts and vectors, the transition parameters were defined as temperature-dependent development rates.

For humans, the intrinsic incubation process was modeled through $b_{3H} = \frac{1}{\tau_H}$, where τ_H denotes the intrinsic incubation period.

For mosquitoes, the constant transition parameter $b_3 = 1/\tau_M$ from the original formulation was replaced by a Brière-type function,

$$b_{3M}(T) = \eta T(T - T_{\min})(T_{\max} - T)^{1/2}, \tag{4}$$

which incorporates minimum and maximum thermal thresholds for sporogonic development [24, 25].

The scaling coefficient η was estimated by imposing $b_{3M}(T_{\text{opt}}) = \frac{1}{\tau_M(T_{\text{opt}})}$, yielding

$$\eta = \frac{\frac{1}{\tau_M(T_{\text{opt}})}}{T_{\text{opt}}(T_{\text{opt}} - T_{\min})(T_{\max} - T_{\text{opt}})^{1/2}} \tag{5}$$

Another addition was the inclusion of a humidity-dependent survival component $p(H)$ on to the daily survival rate of the adult mosquitoes, which is modeled as a sigmoid function of the daily minimum relative humidity H. The parameter H_0 (in %) represents the humidity threshold at which the mosquito experiences a 50% reduction in survival relative to its maximum, and k (in $\%^{-1}$) controls the steepness of the transition. The parameter φ (dimensionless) establishes a baseline survival floor, accounting for the persistence of individuals in favorable micro-habitats even under extreme desiccation conditions.

The SEIRS-SEI complete model is then given by:

$$\frac{dS_H}{dt} = -\lambda_H(t)S_H + \omega R_H \tag{6a}$$
$$\frac{dE_H}{dt} = \lambda_H(t)S_H - b_{3H}E_H \tag{6b}$$
$$\frac{dI_H}{dt} = b_{3H}E_H - \gamma I_H \tag{6c}$$
$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \tag{6d}$$
$$\frac{dS_M}{dt} = \Lambda(t) - \lambda_M(t)S_M - \mu S_M \tag{6e}$$
$$\frac{dE_M}{dt} = \lambda_M(t)S_M - (\mu + b_{3M}\ell)E_M \tag{6f}$$
$$\frac{dI_M}{dt} = b_{3M}\ell E_M - \mu I_M \tag{6g}$$

The forces of infection are defined as:

$$\lambda_H(t) = a\, b_2 \frac{I_M(t)}{N(t)} \tag{7a}$$
$$\lambda_M(t) = a\, b_1 \frac{I_H(t)}{N(t)} \tag{7b}$$

2.6. Recalibration of the temperature-dependent biting rate

In the original formulation by Parham & Michael [11], the biting rate is expressed as a temperature-dependent function derived from the duration of the gonotrophic cycle. Although this formulation captures the general thermal dependence of mosquito feeding behavior, its parameterization was not developed specifically for Amazonian mosquito populations. With this in mind, we recalibrated the biting-rate function using entomological observations reported by de Barros, Honório and Arruda [26], regarding females of *Anopheles darlingi* collected between August 2003 and July 2004 through intermittent sampling carried out in selected months over the study period.

Their results allowed for estimation of the duration of the gonotrophic cycle through the following formula:

$$P = Q^{1/g} \Longrightarrow g = \frac{\ln(Q)}{\ln(P)}, \tag{8}$$

where P is the daily survival rate of mosquitoes, Q is the proportion of parous mosquitoes females in the population and g is the duration of the gonotrophic cycle, given in days [27].

Given the lack of more recent data, it was decided to use the obtained estimates of the time period specified above with average temperatures from the period of analysis between 2017 and 2023, given that the biting rate is given as a function of temperature. It was decided to consider $T' = 16°C$, given that below this temperature there is no development of the vector, as the threshold temperature for the biting rate, and we estimated values of g for the missing months of the analysis via interpolation. With this, given the closed formula for the biting rate, and knowing that it is the inverse of the gonotrophic cycle, we calculated $D_1 = g(T - T')$ for each of the months with available data.

Having monthly values of D_1, we took their average in

order to estimate the biting rate and the proportion of female mosquitoes to humans, which is given by

$$m = \frac{HBR}{a(T)}, \tag{9}$$

where HBR is the mean biting density of *An. darlingi*, given in bites/person/night, while $a(T)$ is the biting rate, given in bites/mosquito/night. Based on the data, M_0, the initial population of mosquitoes used in the analysis, was estimated to be approximately $10N_0$, based on the found value of m for the month of January.

At this point, we also approximated the ratio of initially exposed to infectious mosquitoes to be their values at the steady state:

$$\frac{dI_M}{dt} = b_{3M}\ell E_M - \mu I_M = 0$$
$$\implies \frac{E(0)}{I(0)} = \frac{\mu(0,0)}{b_{3M}(0)\ell(0,0)} \approx 3$$

This adjustment preserves the functional structure proposed by Parham and Michael while adapting the parameter values to better reflect the feeding behavior of the primary malaria vector in the Amazon region.

2.7. Density-dependent mosquito recruitment and carrying capacity

Mosquito population dynamics were initially modeled using a linear birth–death process of the form

$$\frac{dM}{dt} = b - \mu M, \tag{10}$$

where b represents the birth rate and μ is the per capita mortality rate. However, this formulation does not account for environmental limitations on population growth.

To incorporate the effect of limited environmental resources, we introduce a density-dependent recruitment mechanism inspired by the classical Verhulst logistic model [28],

$$\frac{dM}{dt} = bM\left(1 - \frac{M}{K}\right) - \mu M, \tag{11}$$

where $M(t) = S_M(t) + E_M(t) + I_M(t)$ denotes the total mosquito population, b the per capita recruitment rate, μ the mortality rate, and K the environmental carrying capacity. This formulation captures the reduction in population growth as the mosquito population approaches the environmental limit.

In the present model, this idea is adapted by replacing the constant recruitment term with the density-dependent function

$$\Lambda(t) = b(R,T)\,K(t)\left(1 - \frac{M(t)}{K(t)}\right), \tag{12}$$

where the environmental dependence is separated into two distinct mechanisms:

1. The climatic modulation of mosquito recruitment through $b(R,T)$, inherited from the original Parham-Michael formulation;
2. The environmental limitation imposed by habitat availability through the carrying capacity $K(t)$.

In this interpretation, $b(R,T)$ governs the rate at which mosquitoes are produced under favorable temperature and rainfall conditions, whereas $K(t)$ determines the maximum sustainable mosquito population supported by the environment. Consequently, temperature and precipitation affect mosquito dynamics both through reproductive success and through the availability of suitable breeding habitats.

The environmental carrying capacity is defined by

$$K(t) = M' \cdot f_T(T(t)) \cdot f_R(R(t)), \tag{13}$$

where M' denotes the baseline mosquito population size. Given that we require $0 \le 1 - \frac{M(t)}{K(t)} \le 1$, we would require M' such that $M' \ge \frac{M}{f_T(T)f_R(R)}$.

The environmental response functions are given by

$$f_T(T) = \min\left(1.0, \frac{T(T - T_{\text{min A.}})\sqrt{T_{\text{max A.}} - T}}{T_{\text{opA.}}(T_{\text{opA.}} - T_{\text{min A.}})\sqrt{T_{\text{max A.}} - T_{\text{opA.}}}}\right), \tag{14a}$$

$$f_R(R) = \left(\frac{R}{R_L/2}\right)\exp\left(1 - \frac{R}{R_L/2}\right), \tag{14b}$$

where $f_T(T)$ describes the temperature suitability for mosquito persistence and $f_R(R)$ represents the dual effect of rainfall on habitat availability.

The rainfall response function increases with precipitation under moderate rainfall conditions, reflecting the formation of breeding sites, but decreases under excessive rainfall due to habitat flushing and destruction of immature stages.

Both environmental response functions are normalized so that $0 \le f_T(T), f_R(R) \le 1$, ensuring that the carrying capacity remains bounded by the baseline population scale M_0.

Unlike the classical logistic formulation, where population growth is directly proportional to the current mosquito abundance, the present representation assumes that recruitment is primarily determined by environmental suitability and breeding-site availability. The factor $\left(1 - \frac{M(t)}{K(t)}\right)$ therefore imposes density-dependent regulation by reducing recruitment as the mosquito population approaches the carrying capacity, providing a biologically realistic upper bound on mosquito abundance. The mathematical implications of this formulation are analyzed in Section 3.1.

2.8. Objective function and parameter bounds

With the model structure, data, and environmental predictors defined, we describe the computational procedures

used to calibrate the model to the observed case data, including the objective function, the calibration strategy, and the numerical implementation. Model parameters were estimated by minimizing the discrepancy between the simulated infected-human trajectory $I_H(t)$ and the observed active-case counts reported through SIVEP-Malaria, restricted to rural Manaus (scaled by the rural share of reported cases between 2015 and 2019, which was found to be 41.54%). The objective function is a weighted mean squared error (wMSE):

$$\mathrm{wMSE}(\theta) = \frac{\sum_{i=1}^{n} w_i \left(I_H(t_i;\theta) - y_i\right)^2}{\sum_{i=1}^{n} w_i}, \tag{15}$$

where y_i denotes the observed active-case count at date t_i, $I_H(t_i;\theta)$ is the model prediction obtained by linear interpolation of the simulated trajectory onto the observation dates, and θ is the vector of calibrated parameters.

The weights w_i follow an exponential decay that prioritizes early-time dynamics within each year:

$$w_i = w_0 \exp\left(-\lambda\, d_i\right), \tag{16}$$

where d_i is the number of days from the start of the year for observation i, $w_0 = 5.0$ is the initial weight, and $\lambda = 0.0015$ day^{-1} is the decay rate. This weighting scheme reflects the greater sensitivity of the early-season trajectory to initial conditions and parameter values, guiding the optimizer toward solutions that correctly capture the onset of the rainy-season transmission peak.

An extinction penalty was added to the objective to prevent the optimizer from collapsing the infected-mosquito population to near-zero levels, which would sever the force-of-infection coupling and render the human transmission parameters structurally unidentifiable:

$$J(\theta) = \mathrm{wMSE}(\theta) + \mathcal{P}\left(I_M^{\min}(\theta)\right), \tag{17}$$

where $I_M^{\min}(\theta) = \min_t I_M(t;\theta)$ is the annual minimum of the infected-mosquito compartment, and

$$\mathcal{P}\left(I_M^{\min}\right) = \begin{cases} 0 & \text{if } I_M^{\min} \geq I_M^*, \\ \mathcal{B}\left(1 + \dfrac{I_M^* - I_M^{\min}}{I_M^*}\right) & \text{if } I_M^{\min} < I_M^*, \end{cases} \tag{18}$$

with $I_M^* = 10$ (viability threshold) and $\mathcal{B} = 10^8$ (penalty base scale). The penalty scales linearly with the depth of extinction, ensuring that partially-extinct trajectories are ranked proportionally rather than receiving a uniform penalty that would flatten the objective landscape.

The nine calibrated parameters were grouped into four biological "groups" for sequential fitting:

1. **Humidity response** (H_0, k, φ): the half-saturation threshold, sigmoid steepness, and baseline survival floor of the humidity-dependent survival function $p(H)$;
2. **Human transmission** (τ_H, γ, ω): the intrinsic incubation period, recovery rate, and immunity waning rate;
3. **Force of infection** (b_1, b_2): the human-to-mosquito and mosquito-to-human transmission probabilities;
4. **Carrying capacity** (M'): the baseline mosquito population scale that determines the environmental carrying capacity $K(t)$.

Table 1
Parameter bounds for sequential topic calibration.

Parameter	Lower bound	Upper bound
H_0 (%)	20	75
k (%$^{-1}$)	0.01	1.0
φ	0.01	0.5
τ_H (days)	10	20
γ (days^{-1})	1/150	1.0
ω (days^{-1})	1/300	0.1
b_1	0.001	0.5
b_2	0.01	0.5
M'	50,000	$500 \times N_y$ [1]

[1] N_y denotes the estimated rural population of Manaus for year y.

The parameter bounds are given in Table 1. The lower bound on $H_0 = 20\%$ prevents the humidity response from collapsing (which would drive $\mu \to \infty$ and extinguish the vector population). The bound on $b_2 \geq 0.01$ ensures a minimum level of mosquito-to-human transmission, while $\tau_H \in [10, 20]$ days constrains the human incubation period to the biologically plausible range for *P. vivax* [29].

2.9. Year-by-year sequential calibration

Parameters were calibrated sequentially for each year from 2017 to 2023 using the `lmfit` implementation of differential evolution (DE) [30]. The DE settings were: population multiplier `popsize` = 6, maximum function evaluations `max_nfev` = 800, convergence tolerance `tol` = 0.01, and random seed fixed at 42 for reproducibility.

Within each year, the four groups were fitted in the fixed order:

$$\underbrace{(H_0,\, k,\, \varphi)}_{\text{humidity}} \to \underbrace{(\tau_H,\, \gamma,\, \omega)}_{\text{human}} \to \underbrace{(b_1,\, b_2)}_{\text{FoI}} \to \underbrace{M'}_{\text{capacity}} \tag{19}$$

At each step, only the current topic's parameters were free; all previously fitted parameters were held fixed at their most recent estimates. This sequential strategy reduces the dimensionality of each optimization sub-problem while respecting the biological ordering: the humidity response determines mosquito mortality, which constrains the infection cycle, which in turn determines the force of infection and carrying capacity.

After the sequential pass, a joint refinement was performed using the Nelder-Mead simplex method [31], with all nine parameters free simultaneously, seeded from the sequential solution. A seed-evaluation guard was applied: after each DE run and after each refinement stage, the starting point was explicitly evaluated, and the optimizer's output was accepted only if it improved upon the seed. This prevents the stochastic DE from returning a solution worse than its initialization.

The model was calibrated sequentially across years, with the end-state vector of each year serving as the initial state for the next:

$$\mathbf{z}_0^{(y+1)} = \mathbf{z}_{\text{end}}^{(y)}, \tag{20}$$

where $\mathbf{z} = (S_H, E_H, I_H, R_H, S_M, E_M, I_M)$ is the full state vector. The human infected compartment was always reset to the first observed active-case count for the new year:

$$I_{H\,0}^{(y+1)} = \text{round}\left(y_{\text{first}}^{\text{obs}}\right), \tag{21}$$

while all other compartments were carried forward without modification. Negative end-state values were clamped to zero before carrying.

For the first year (2017), initial-condition fractions were explicitly fitted using Nelder-Mead with 300 function evaluations:

$$S_{H,0} = \alpha_S \cdot N, \quad E_{H,0} = \alpha_E \cdot N, \quad I_{M,0} = \alpha_I \cdot M_0, \tag{22}$$

where $\alpha_S \in [0.3, 0.9]$, $\alpha_E \in [0.001, 0.30]$, and $\alpha_I \in [0.0005, 0.15]$, with the remaining compartments determined by conservation: $R_{H,0} = N - S_{H,0} - E_{H,0} - I_{H,0}$, $M_0 = 10N$, and $S_{M,0} = M_0 - E_{M,0} - I_{M,0}$.

2.10. Weekly beta diagnostic

To evaluate where the climate-driven formulation deviates from what the data demands, a diagnostic calibration was performed in which the effective transmission coefficients were freed as per-week parameters instead of being derived from the biting rate and fixed b_1, b_2:

$$\lambda_H(t) = \beta_{h,w}\,\frac{I_M(t)}{N}, \qquad \lambda_M(t) = \beta_{m,w}\,\frac{I_H(t)}{N}, \tag{23}$$

where $w = \lfloor d/7 \rfloor$ is the week index and d is the day number within the year. This yields $2{\times}n_{\text{weeks}}$ free parameters per year (typically 106 for a 365-day year), with bounds $\beta_{h,w}, \beta_{m,w} \in [10^{-9}, 3.0]$.

All other model parameters were fixed at the values obtained from the yearly topic calibration. The initial state was reconstructed from the previous year's end-state, with I_H reset to observed data. The optimizer used was `lmfit.minimize` with the trust-region reflective method (`least_squares`), with `max_nfev` $= 1200$ and `ftol` $= 10^{-8}$.

The residual vector for the trust-region solver was constructed as:

$$r_i = \sqrt{w_i}\,\left(I_H(t_i) - y_i\right) + \sqrt{\mathcal{P}\left(I_M^{\min}\right)}, \tag{24}$$

where $\sqrt{\mathcal{P}}$ is injected as a constant offset to preserve the residual vector length while providing a gradient that pushes I_M upward when the mosquito population approaches the viability threshold.

The weekly beta fit serves as a diagnostic baseline: it reveals the best achievable fit when the transmission coefficients are unconstrained by the climate formulation. Persistent deviations between the weekly-optimal $\beta_{h,w}$ and the climate-driven $a(T) \cdot b_2$ indicate periods where the environmental model fails to capture the true transmission dynamics.

There are three main observations that allow us to use the effective transmission coefficients $\beta_{h,w}$ and $\beta_{m,w}$ as per-week free parameters:

(i) The transmission coefficients are intrinsically time-varying: In the mechanistic formulation, the forces of infection are given by $\lambda_H = a(T)\,b_2\,I_M/N$ and $\lambda_M = a(T)\,b_1\,I_H/N$, where the biting rate $a(T)$ is the only environmental factor and the per-bite probabilities b_1 and b_2 are assumed to be constants. This assumption is biologically restrictive because these coefficients summarize several processes that may vary over the transmission season, including parasite development within the vector, vector susceptibility to infection, vector to host contact, and host infectiousness. These processes can be affected by temperature, rainfall, humidity, and changes in vector and host demography [24, 32]. We therefore relax the assumption that b_1 and b_2 remain constant throughout the study period and allow the data to determine how transmission changes over time.

(ii) The weekly fit provides a nonparametric estimate of the latent transmission process: We represent the effective transmission coefficients as latent time-varying functions, $\beta_h(t)$ and $\beta_m(t)$. These functions are approximated as piecewise constant over weekly intervals:

$$\beta_h(t) = \beta_{h,w},\ \beta_m(t) = \beta_{m,w},\ t \in [7w, 7(w+1))$$

This provides a flexible finite-dimensional representation of possible transmission profiles. The weekly coefficients are estimated by minimizing the weighted residual between the model predictions and the observed malaria cases. Thus, rather than imposing a specific relationship between transmission and environmental covariates, the weekly model provides a flexible, data-driven estimate of the temporal variation in transmission.

The resulting $2 \times n_{\text{weeks}}$ coefficients don't represent independent biological parameters to be estimated, but rather provide a flexible empirical representation of the transmission process. The additional degrees of freedom allow the

model to capture temporal changes in transmission that a more constrained model may miss, providing a reference for evaluating the proposed mechanistic model.

(iii) The climate-driven model is nested within the weekly model: The climate-driven model is obtained by imposing the restriction

$$\beta_{h,w} = a(T_w)b_2, \qquad \beta_{m,w} = a(T_w)\, b_1, \tag{25}$$

where b_1 and b_2 are constant over the entire study period. Consequently, the climate-driven formulation is a restricted case of the more flexible weekly model. Because all other model parameters and initial conditions are held fixed, the difference in fit between the two formulations directly measures the extent to which the constant-coefficient assumption limits the model's ability to reproduce the observed cases.

Releasing the coefficients reduces the wMSE by one to two orders of magnitude relative to the climate-driven model. This substantial improvement indicates that the data contain temporal variation in transmission that cannot be adequately represented by constant b_1 and b_2.

The weekly estimates are therefore used as a diagnostic benchmark rather than as the final parameterization of the model. They identify the temporal transmission pattern that a parsimonious mechanistic formulation should seek to explain. This approach preserves the distinction between estimating a flexible empirical transmission process and identifying the underlying biological parameters, as the weekly coefficients characterize the observed transmission signal, while the subsequent structural model determines how that signal can be explained by environmental drivers.

Finally, the weekly estimates provide the link between the empirical transmission signal and the environmental model. Since the biting rate $a(T_w)$ is explicitly modeled as a function of temperature, the corresponding weekly per-bite transmission probabilities can be recovered as

$$b_{1,w} = \frac{\beta_{m,w}}{a(T_w)}, \qquad b_{2,w} = \frac{\beta_{h,w}}{a(T_w)}. \tag{26}$$

These time-varying estimates represent the per-bite transmission probabilities implied by the observed case dynamics after accounting for the temperature-dependent biting rate. They are subsequently used as empirical targets for regression models relating b_1 and b_2 to environmental drivers. In this way, the weekly fit serves as an intermediate step in the modeling chain

$$\text{environment} \longrightarrow (b_1, b_2) \longrightarrow \text{transmission} \longrightarrow \text{cases},$$

allowing a flexible, data-driven estimate of transmission to be translated into a simplified and mechanistically interpretable environmental model.

2.11. Numerical solver and implementation

All ODE simulations were performed using Python version 3.12.7 (conda-forge distribution), using the LSODA solver as implemented in `scipy.integrate.solve_ivp` (SciPy version 1.17.1), with automatic stiffness detection and order switching. The integration was evaluated on a fine grid of 10 points per day ($\Delta t = 0.1$ day), with the solution interpolated back to daily resolution via linear interpolation. End-state vectors were clamped to zero to prevent negative compartment values from propagating through the year chain.

Climate inputs (temperature, precipitation, minimum humidity) were smoothed using a 14-day rolling mean for mosquito demographic rates (mortality μ, sporogonic development b_{3M}, birth rate Λ), while the biting rate $a(T)$ used raw daily temperature to preserve high-frequency transmission variability.

Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

All computations were performed on a laptop running Windows 11 (64-bit) equipped with a 12th Gen Intel Core i5-1235U processor (10 cores, 12 threads) and 16 GB RAM.

3. Mathematical Analysis of the Model

3.1. Mosquito population dynamics

In this section, we analyze the qualitative impact of the density-dependent recruitment term on mosquito population dynamics. We compare the proposed formulation with the classical Verhulst model and show that the modified structure prevents extinction under suitable environmental conditions.

Proposition 1 (Environmental persistence of the mosquito population)**.** *Assume that $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$ for all $t \geq 0$. Then, the mosquito population $M(t)$ governed by*

$$\frac{dM}{dt} = b(R,T)\, K(t) \left(1 - \frac{M(t)}{K(t)}\right) - \mu M(t) \tag{27}$$

cannot converge to zero as $t \to \infty$. In particular, $M = 0$ is not an equilibrium and cannot be asymptotically stable.

Proof. We begin by recalling the classical logistic model with mortality:

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K}\right) - \mu M. \tag{28}$$

In this formulation, the right-hand side vanishes at $M = 0$, so that $M = 0$ is always an equilibrium. Depending on the parameters, this equilibrium may be stable, allowing extinction of the population.

In contrast, in the present model the recruitment term is given by

$$\Lambda(t) = b(R,T)\, K(t) \left(1 - \frac{M(t)}{K(t)}\right), \tag{29}$$

and the population dynamics satisfy

$$\frac{dM}{dt} = b(R,T)\,K(t)\left(1-\frac{M(t)}{K(t)}\right) - \mu M(t). \tag{30}$$

Evaluating the right-hand side at $M = 0$, we obtain

$$\left.\frac{dM}{dt}\right|_{M=0} = b(R,T)\,K(t). \tag{31}$$

By assumption, $b(R(t),T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$, hence

$$\left.\frac{dM}{dt}\right|_{M=0} \geq b_0 K_0 > 0. \tag{32}$$

Therefore, $M = 0$ is not an equilibrium of the system.

Moreover, for $M(t)$ sufficiently small, we have

$$\frac{dM}{dt} \geq b_0 K_0 - \mu M(t). \tag{33}$$

Choosing $\varepsilon > 0$ such that $\varepsilon < \frac{b_0 K_0}{\mu}$, it follows that

$$\frac{dM}{dt} > 0 \quad \text{whenever } 0 < M(t) < \varepsilon.$$

Thus, any trajectory entering a neighborhood of zero is driven away from it.

Consequently, $M(t)$ cannot converge to zero as $t \to \infty$, and the extinction state is not asymptotically stable. □

3.2. Basic Reproduction Number ($\mathcal{R}_0$)

We obtain $\mathcal{R}_0$ via the next-generation matrix method [33], ordering the infected compartments as

$$X_I = (E_H, I_H, E_M, I_M)^\top$$

and writing the model as $\dot{x} = \mathcal{F}(x) - \mathcal{V}(x)$. At the disease-free equilibrium $S_H^* = N$, $S_M^* = M^*$, so

$$x_0 = (N, 0, 0, 0, M^*, 0, 0)^\top, \tag{34}$$

with equilibrium vector abundance and recruitment

$$M^* = \frac{\bar{b}\,\bar{K}}{\bar{b}+\mu}, \qquad \Lambda^* := \Lambda\Big|_{M=M^*} = \mu M^* = \frac{\mu \bar{b}\,\bar{K}}{\bar{b}+\mu}, \tag{35}$$

where $\bar{b} = b(\bar{R},\bar{T})$ and $\bar{K}$ are the environmental drivers at a fixed reference state (their study-period means), so that $\mathcal{R}_0$ is a function of that reference climate-habitat state. Here Λ^* is not arbitrary: it is the recruitment that sustains the equilibrium vector population M^* and inherits the environmental dependence of $b(R,T)$ and $K(t)$.

Evaluating the new-infection and transfer matrices at x_0 and taking the spectral radius of FV^{-1} (Appendix B) gives

$$\mathcal{R}_0 = \rho(FV^{-1}) = \sqrt{\frac{a^2 b_1 b_2 \Lambda^* b_{3M}\ell}{N\gamma(b_{3M}\ell+\mu)\mu^2}}, \tag{36}$$

which factors into the classical host-vector form

$$\mathcal{R}_0^H = \frac{a b_2}{\gamma},\ \mathcal{R}_0^M = \frac{a b_1 \Lambda^*}{N\mu}\cdot\frac{b_{3M}\ell}{(b_{3M}\ell+\mu)\mu}, \qquad \mathcal{R}_0 = \sqrt{\mathcal{R}_0^H\,\mathcal{R}_0^M}, \tag{37}$$

where $\mathcal{R}_0^H$ is the expected number of humans infected by a single infectious mosquito and $\mathcal{R}_0^M$ the expected number of mosquitoes that survive the latent period and become infectious, generated by a single infectious human. The geometric mean arises from the two-generation host-vector cycle: the infection must complete a full human→mosquito→human loop, which the block structure of FV^{-1} decouples into these two factors.

$\mathcal{R}_0$ is independent of both the intrinsic incubation period τ_H and the waning rate ω: the former because the model has no human disease-induced mortality (every exposed individual eventually becomes infectious, so the latent period affects only the timing of secondary infections), and the latter because no individuals are immune at the disease-free equilibrium, so waning is irrelevant to the linearized invasion dynamics.

3.3. Sensitivity analysis methodology

The reproduction number in Eq. (37) depends on the reference habitat state through the mosquito recruitment $\Lambda^* = \mu M^*$, which inherits the environmental dependence of the recruitment function $b(\bar{R},\bar{T})$ and of the carrying capacity $\bar{K}$ (Eq. 35). The calibrated parameters and the environmental drivers therefore enter $\mathcal{R}_0$ nonlinearly and in a mutually coupled way. To quantify how accurately each of them must be estimated we report the normalized sensitivity index of Abdelrazec and Gumel [28],

$$S_x = \frac{x}{Y}\frac{\partial Y}{\partial x} = \frac{\partial \log Y}{\partial \log x}, \tag{38}$$

where Y is an output of the model and x one of its inputs. S_x is the relative change of Y produced by a 1% relative change of x: a large $|S_x|$ means that a small relative error on x is amplified into a large relative error on Y, so that parameter deserves a careful estimate, whereas $S_x = 0$ means that the output is locally insensitive to x. The index is dimensionless, and being a logarithmic derivative it is directly comparable across parameters with different units. Because each output is decomposed about a single physically meaningful operating point rather than over a sampled hypercube, no probability law is assumed on the inputs, which are admissible biological intervals; the price is that interactions between parameters are then resolved

only through the curvature of the response away from that point, the usual trade-off between local and global sensitivity methods [40].

Two outputs are decomposed, each about its own reference point. The first is the closed-form reproduction number of Eq. (37), evaluated at a single reference environmental state: the mean climate of the transmission season, that is, of the days of 2017-2023 on which the model sustains $\mathcal{R}_0 > 1$ (39.8% of the study-period days, with $\bar{T} = 25.57\,°\mathrm{C}$, $\bar{R} = 7.19\,\mathrm{mm}$ and $\bar{H} = 78.34\%$). Averaging over the transmission season rather than over all days avoids evaluating the bell-shaped rainfall-recruitment response at an arithmetic-mean rainfall dominated by dry days, which would force $\mathcal{R}_0$ below the threshold. At this reference the pre-calibration parameter values of Table A.1 give $\mathcal{R}_0 = 2.20$, with $\mathcal{R}_0^H = 4.17$ and $\mathcal{R}_0^M = 1.16$, the carrying-capacity scale being $M' = 468{,}580$.

Because $\mathcal{R}_0$ is available in closed form, its index is obtained analytically. The elementary response functions of Table A.1 are assembled symbolically into $\mathcal{R}_0$ and Eq. (38) is differentiated by the chain rule over the composite rates of the model (the biting rate a, the mortality μ, the sporogonic development rate b_{3M}, the sporozoite survival ℓ, the development times τ_M and τ_L, the recruitment rate $\bar{b}$ and the recruitment Λ^*) which are themselves functions of the primary parameters. As an independent check the same index is also evaluated numerically, by central differences of the implemented closed form,

$$S_x \approx [Y(x(1+h)) - Y(x(1-h))]/[2hY],$$

with $h = 0.05$. The two agree to 2.6×10^{-3} over the parameters for which the index is a smooth function of its input, the residual being the $O(h^2)$ truncation error of the difference quotient (verified to decrease quadratically as h is reduced from 0.05 to 0.005).

The second output is read from the numerical solution of Section 2.11 and therefore retains the environmental forcing and the nonlinear couplings that the closed form suppresses: the seasonal maximum of the infectious-human compartment,

$$Q_{\mathrm{peak}}(\theta) = \max_t I_H(t;\theta), \tag{39}$$

averaged over the calibrated years 2017-2023, which amounts to $Q_{\mathrm{peak}} = 1{,}033$ infectious individuals. No closed form exists for this output, so its index is *estimated* with the same central difference ($h = 0.05$), each perturbation re-integrating all seven years; the nine calibrated parameters are perturbed about their per-year calibrated value and the sixteen climate-response constants about the baseline values of Tables A.1 and A.2.

The model carries more parameters than the 25 decomposed here. Two entries of Table A.2 are not free parameters at all: $b_{3H} = 1/\tau_H$ and $b_{3M}(T) = \eta\, T(T - T_{\min})(T_{\max} - T)^{1/2}$, the latter a function of the sporogonic constants already in the list, its scaling coefficient η being fixed by the constraint $b_{3M}(T_{\mathrm{opt}}) = 1/\tau_M(T_{\mathrm{opt}})$; perturbing $(DD, T^{\mathrm{p}}_{\min}, T^{\mathrm{p}}_{\max}, T^{\mathrm{p}}_{\mathrm{opt}})$ therefore already propagates to b_{3M}. Of the 30 primary parameters of Table A.2, five, which are the three maximum survival probabilities (p_{ML}, p_{MP}, p_{ME}) and the two stage durations (τ_E, τ_P), are held at their literature values in every simulation and calibration and are not perturbed. The remaining 25 are the ones decomposed below, and the admissible range of each, that is, the box sampled by the global analysis of Section 4.4, is given in Table A.3.

Those 25 are reported in eight groups defined by the mechanism each parameter acts through, which is the taxonomy used in Table 3 and in the bar colours of Figs. 4 and 5: the per-bite transmission probabilities together with the capacity scale (b_1, b_2, M'); the human immunity and turnover rates (τ_H, γ, ω); the humidity-dependent daily adult survival (H_0, k, φ); the biting rate (T', D_1); the sporogonic development $(DD, T^{\mathrm{p}}_{\min}, T^{\mathrm{p}}_{\max}, T^{\mathrm{p}}_{\mathrm{opt}})$; the immature-stage survival and rainfall response, which together set the recruitment rate (R_L, B_E, c_1, c_2); the adult survival curve, and hence the mortality (A, B, C); and the *Anopheles* thermal suitability of the carrying capacity $(T^{\mathrm{a}}_{\min}, T^{\mathrm{a}}_{\mathrm{opt}}, T^{\mathrm{a}}_{\max})$. Grouping by mechanism rather than by calibration status is what keeps the groups interpretable: H_0, k, φ belong together although they are calibrated, τ_H, γ, ω belong together although only γ enters $\mathcal{R}_0$, and no group mixes a biting-rate threshold with a parasite development optimum.

The superscripts p and a distinguish the *Plasmodium* thresholds $T_{\min}, T_{\max}, T_{\mathrm{opt}}$ from the *Anopheles* ones of Table A.2; note that the biting-rate threshold T' and the vector development threshold $T^{\mathrm{a}}_{\min}$ share the baseline value 16 °C but are distinct parameters of distinct response functions, and are perturbed independently.

Three further cases require care. The two parameters that cancel from the closed form of Section 3.2, τ_H and ω, have analytic index exactly zero, a property of the next-generation matrix and not an empirical finding, whereas in the integrated dynamics they are estimated like any other parameter, so that the analytical and the numerical statements can be compared. The three *Anopheles* suitability temperatures act on $\mathcal{R}_0$ only through the thermal suitability factor f_T, which is saturated ($f_T = 1$) at the reference climate; their analytic index is therefore zero there, and a finite-difference estimate that crosses the clamp is not meaningful, so they are excluded from the analytic-versus-numeric comparison. Finally, the population size N is not a free parameter of the calibration but carries a constant index $-1/2$, and is tabulated for completeness. Every index of Table 3 is produced by a single script from a fixed reference point, with no sampling, so that the table and Fig. 4 are exactly reproducible; the global analysis of Section 4.4 samples the box of Table A.3 instead.

Table 2
Year-by-year calibrated parameters and weighted mean squared error (wMSE) of the SEIRS-SEI model, 2017-2023.

Year	H_0	k	φ	τ_H	γ	ω	b_1	b_2	M'	wMSE
2017	37.58	0.705	0.215	19.81	0.0071	0.0855	0.259	0.035	50,723	85,036
2018	20.36	0.046	0.490	20.00	0.0073	0.0961	0.487	0.022	51,245	64,959
2019	22.13	0.053	0.493	19.96	0.0072	0.0744	0.488	0.022	50,474	41,436
2020	20.13	0.039	0.500	19.98	0.0067	0.0422	0.500	0.019	50,024	23,541
2021	20.02	0.043	0.493	19.99	0.0067	0.0773	0.500	0.019	50,015	39,449
2022	20.02	0.043	0.493	19.77	0.0099	0.0312	0.500	0.019	50,832	25,046
2023	20.14	0.207	0.431	19.86	0.0081	0.0310	0.485	0.010	53,616	18,360

4. Results

4.1. Recalibration of the temperature-dependent biting rate

The temperature-dependent biting rate was recalibrated using field estimates of *Anopheles darlingi* feeding behavior, as described in Section 2.6. The recalibrated function has a minimum biting temperature threshold of $T' = 16°C$ and a thermal scaling factor $D_1 = 24.8°C$ days. Compared with the original Parham & Michael parameterization, the recalibrated function better reflects the temperature range observed in the Amazon and produces higher biting rates during the transmission season in Manaus. This recalibrated biting rate is used in all subsequent simulations and in the weekly transmission regressions presented in Section 4.6.

4.2. Model calibration reproduces the observed transmission dynamics

The SEIRS-SEI model was calibrated year by year over 2017-2023 following the sequential topic scheme of Section 2.9, and the recovered parameters are summarized in Table 2. Figure 2 shows the fitted infectious compartment I_H against the observed active-case count alongside the weekly free-beta fit for 2017, which exhibited the largest relative improvement of the weekly model over the constant-coefficient formulation (Section 4.5). The calibrated parameters remain within their physiologically prescribed bounds found in Table 1, with the humidity response (H_0, k, φ) and the human recovery parameters (τ_H, γ, ω) showing limited cross-year variation, while the force-of-infection coefficients b_1 and b_2 consistently approach their upper and lower bounds, respectively (Table 2).

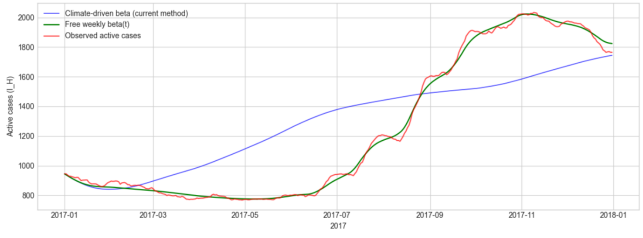


Figure 2: Comparison between the observed active cases and the model estimation of the infectious human compartment I_H for 2017.

Figure 3 shows the seasonal variation of both human and coupled $\mathcal{R}(t)$ during 2017, obtained by substituting the calibrated parameters and the observed drivers into Eq. (37). The reproduction number follows the annual transmission cycle, exceeding unity during the wet season and dropping below the epidemic threshold in the months of lowest precipitation. Because $\Lambda^* = \mu M^*$ inherits the biting rate and carrying capacity as functions of the environmental drivers, this curve makes explicit the dependence of the invasion threshold on the climate-habitat state highlighted in Section 3.2, and motivates the week-specific regression of the transmission probabilities pursued in Section 4.6.

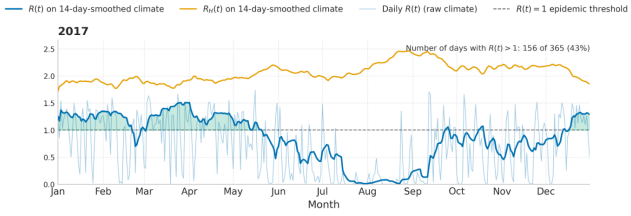


Figure 3: Seasonal variation of the coupled and human basic reproduction number $\mathcal{R}(t)$ for the calibrated model in 2017.

4.3. Local sensitivity of the invasion threshold and of the seasonal peak

For the closed-form reproduction number the normalized index of Eq. (38) is available analytically, and the chain rule over the composite rates of Eq. 35 resolves it exactly. Writing μ for the mortality rate, a for the biting rate, b_{3M} for the sporogonic development rate, ℓ for the sporozoite survival probability, τ_M for the extrinsic incubation period, $\bar{b}$ for the recruitment rate and Λ^* for the equilibrium recruitment, the indices that do not depend on the values of the intermediate rates are

$$S_a = 1,\ S_{b_1} = S_{b_2} = S_{\Lambda^*} = \frac{1}{2},\ S_\gamma = S_N = -\frac{1}{2},\ S_{D_1} = -1, \tag{40}$$

together with

$$\begin{aligned} S_{T'} &= -\frac{T'}{\bar{T} - T'} = -1.67, \\ S_{b_{3M}} = S_\ell &= \frac{\mu}{2(b_{3M}\ell + \mu)} = +0.39, \end{aligned} \tag{41}$$

and, for the mortality rate that couples every other channel,

$$S_\mu = \frac{1}{2}\left[\frac{\bar{b}}{\bar{b}+\mu} - 2 - \frac{b_{3M}\ell\,(\mu\tau_M + 1)}{b_{3M}\ell + \mu}\right] = -1.39. \quad (42)$$

The remaining indices are the values of Table 3, obtained by the same symbolic differentiation of the assembled closed form. Two structural readings follow. First, the two transmission probabilities and the recruitment Λ^* enter $\mathcal{R}_0$ only through the square root of Eq. 37, hence the universal index $1/2$: a 10% error on either probability shifts the invasion threshold by only 5%. Second, the thermal-suitability factor f_T is saturated at the reference climate, so the three *Anopheles* suitability temperatures have index exactly zero there, as they matter only for carrying capacity away from the reference state.

With the exception of H_0, every index of Table 3 is dominated by the mosquito vital rates. The linear coefficient B of the adult survival curve is the single most influential quantity ($S_B = +4.58$), followed by its quadratic coefficient A (-2.68), the biting-rate threshold T' (-1.67), and D_1 and $T_{\min}$ (each ≈ -1.04). The opposite signs of A and B are not a contradiction: what enters the threshold is the local slope of $p(T)$ at $\bar{T}$, which is positive there, so the invasion condition is governed by the balance of a steeply rising biting rate against a steeply falling vector survival.

The seasonal peak of the calibrated model tells a different story. It is carried by the same two coefficients B ($+3.77$) and A (-2.51), but the maximum temperature of *Plasmodium* development $T^{\mathrm{p}}_{\max}$ now ranks third ($+1.68$) and the biting-rate threshold T' fourth (-1.32). The transmission probabilities b_2 and b_1 fall back to $+0.43$ and $+0.36$, and the recovery rate γ to -0.40. Most strikingly, the humidity response is essentially irrelevant to the height of the wave ($S_{H_0} = -0.004$) although it is one of the leading terms of the invasion threshold ($S_{H_0} = -1.04$): near-saturation of p_H at the reference humidity leaves the peak insensitive to H_0 while μ, and hence $\mathcal{R}_0$, still responds to it.

Figure 4 displays the two decompositions side by side, and Table 3 lists them with the exact analytic values of the reproduction number. In the table, Y is the output being differentiated: the closed-form reproduction number, $Y = \mathcal{R}_0$, in the first column, and the seasonal maximum of the infectious-human compartment, $Y = Q_{\mathrm{peak}}$ of Eq. (39), in the second column. The same S_x therefore denotes two distinct indices, one for each output. The left panel is the finite-difference estimate of the same index; its largest deviation from the analytic value is 0.06 (for H_0) and is purely the $O(h^2)$ truncation error of the difference quotient, so the two representations of S_x agree.

4.4. Global sensitivity of the seasonal peak

The index of Eq. (38) is a local quantity, evaluated at a single reference state. The global sensitivity of the seasonal peak is instead quantified by the variance-based Sobol' decomposition of Q_{peak} of Eq. (39), which apportions the variance of this output over the whole admissible box of the 25 parameters of Table A.3 among the individual parameters and among their interactions [41]. Writing $Y = Q_{\mathrm{peak}}(\mathbf{X})$ for the output as a function of the 25 independent inputs $\mathbf{X} = (X_1, \ldots, X_{25})$ over the box, its variance decomposes

Table 3
Normalized sensitivity index $S_x = \partial \log Y / \partial \log x$ of each decomposed parameter, evaluated at the reference state of Section 3.3. The $\mathcal{R}_0$ column is the exact analytic index; the Q_{peak} column is the central-difference estimate ($h = 0.05$) of the calibrated model averaged over 2017-2023 ($Q_{\mathrm{peak}} = 1{,}033$). Parameters are grouped by mechanism as in Section 3.3. A dash denotes a parameter that is structurally absent from the closed form. The population size N is listed apart from the groups because it is not a free parameter of the calibration.

Parameter	S_x $\mathcal{R}_0$	S_x Q_{peak}
Transmission probabilities and capacity scale		
b_1	+0.500	+0.358
b_2	+0.500	+0.430
M'	+0.500	+0.007
Human immunity and turnover		
τ_H	0	−0.043
γ	−0.500	−0.402
ω	0	+0.014
Humidity-dependent adult survival		
H_0	−1.040	−0.004
k	+0.365	+0.052
φ	+0.004	+0.002
Biting rate		
T'	−1.672	−1.318
D_1	−1.000	−0.848
Sporogonic development (Plasmodium)		
DD	−0.795	−0.318
$T^{\mathrm{p}}_{\min}$	−1.041	−0.846
$T^{\mathrm{p}}_{\max}$	+0.447	+1.680
$T^{\mathrm{p}}_{\mathrm{opt}}$	+0.334	0.000
Immature-stage survival and recruitment		
R_L	−0.436	−0.129
B_E	+0.073	+0.056
c_1	+0.110	+0.074
c_2	−0.052	−0.046
Adult survival curve and mortality		
A	−2.679	−2.514
B	+4.575	+3.775
C	−0.601	−0.498
Anopheles thermal suitability		
$T^{\mathrm{a}}_{\min}$	0	−0.002
$T^{\mathrm{a}}_{\mathrm{opt}}$	0	+0.004
$T^{\mathrm{a}}_{\max}$	0	+0.002
Population size (not a free parameter)		
N	−0.500	–

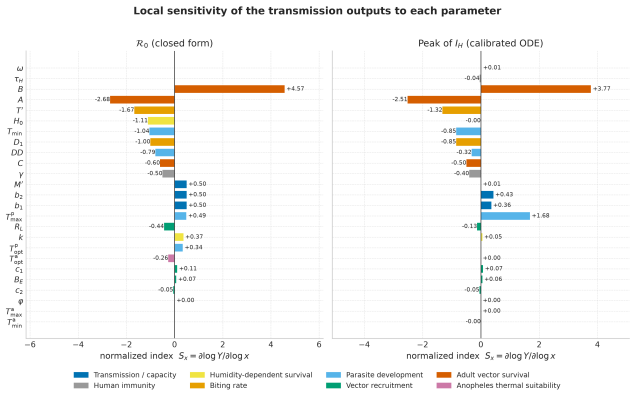


Figure 4: Normalized sensitivity index $S_x = \partial \log Y / \partial \log x$ of the 25 decomposed parameters (bars coloured by parameter group, as in Table 3), where Y is the output being differentiated: $Y = \mathcal{R}_0$ for the closed-form reproduction number in the left panel and $Y = Q_{\text{peak}}$ for the seasonal maximum of the infectious-human compartment of the calibrated model in the right panel. Left: closed-form reproduction number at the reference environmental state, by central differences with $h = 0.05$; the exact analytic values of Table 3 differ from these by at most 0.06, the truncation error of the difference quotient. Right: seasonal maximum of the infectious-human compartment of the calibrated model, averaged over 2017-2023, same scheme. A positive index means the output increases with the parameter. The three *Anopheles* suitability temperatures are exactly zero on the left because the thermal-suitability factor f_T is saturated at the reference climate.

as

$$\mathrm{Var}(Y) = \sum_{i=1}^{25} V_i + \sum_{1 \le i < j \le 25} V_{ij} + \cdots + V_{1\cdots 25}, \tag{43}$$

with the partial variances

$$\begin{aligned} V_i &= \mathrm{Var}_{X_i}\left(\mathbb{E}_{\mathbf{X}_{\sim i}}[Y|X_i]\right), \\ V_{ij} &= \mathrm{Var}_{X_i,X_j}\left(\mathbb{E}_{\mathbf{X}_{\sim ij}}[Y|X_i,X_j]\right) - V_i - V_j, \end{aligned} \tag{44}$$

where $\mathbf{X}_{\sim i}$ denotes all inputs except X_i.

The first-order and total-order Sobol' indices are then

$$\begin{aligned} S_i &= \frac{V_i}{\mathrm{Var}(Y)}, \\ S_{T_i} &= 1 - \frac{\mathrm{Var}_{\mathbf{X}_{\sim i}}\left(\mathbb{E}_{X_i}[Y|\mathbf{X}_{\sim i}]\right)}{\mathrm{Var}(Y)} \\ &= \frac{\mathbb{E}_{\mathbf{X}_{\sim i}}\left[\mathrm{Var}_{X_i}(Y|\mathbf{X}_{\sim i})\right]}{\mathrm{Var}(Y)} \end{aligned} \tag{45}$$

Thus S_i is the fraction of the output variance explained by X_i alone, whereas S_{T_i} is the fraction attributable to X_i together with every interaction in which it takes part; consequently $S_i \le S_{T_i}$, and $\sum_i S_i \le 1$, with equality only in the absence of interactions. The box is sampled with a Saltelli design of $N = 512$ base points, that is $512 \times 27 = 13{,}824$ integrations of the calibrated model, each returning the mean seasonal peak of 2017-2023, and we retain the first-order and total-order indices S_i and S_{T_i} only (Fig. 5).

Because Q_{peak} is a maximum, a parameter set whose transmission is locally subcritical leaves I_H pinned at its initial value, and 74.4% of the draws of the admissible box are of that kind; the decomposition of Fig. 5 is therefore dominated by whether the epidemic takes off at all rather than by the height of an established wave, and its indices describe the box rather than the endemic peak, which is the caveat under which they must be read. Subject to it, the ranking differs sharply from the local one: the recovery rate γ absorbs most of the variance ($S_{T,\gamma} = 0.80$), followed by the curvature A of the adult survival curve (0.25), whereas the linear coefficient B, which dominates Table 3, falls to 0.03, and the three *Anopheles* suitability temperatures stay at the 10^{-5} level because the thermal suitability factor remains saturated over the entire climate range of the study.

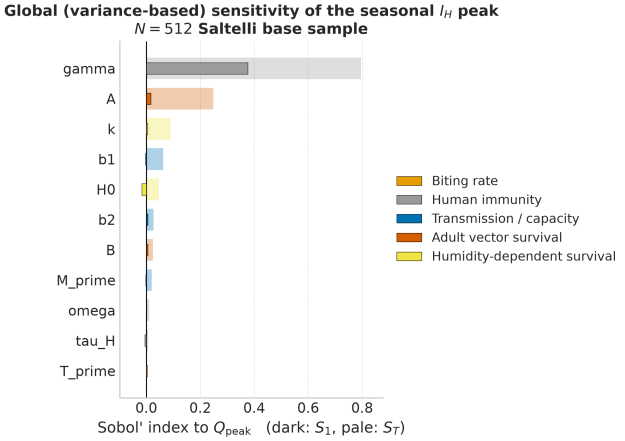


Figure 5: First-order (S_i) and total-order (S_{T_i}) Sobol' indices of the decomposed parameters for the output $Y = Q_{\text{peak}}$, the seasonal peak of the infectious-human compartment defined in Eq. (39), over the whole admissible parameter box of Table A.3 (Saltelli design, $N = 512$; bars coloured by parameter group). Only the 11 parameters with non-negligible sensitivity are shown; the 14 parameters with total-order index below 2.7×10^{-3}, from $T^{\text{p}}_{\text{min}}$ to $T^{\text{p}}_{\text{opt}}$, are omitted from the plot. Each parameter has two overlaid bars: the darker inner bar is the first-order index S_i and the lighter outer bar is the total-order index S_{T_i}. The total-order index also collects the variance that a parameter carries through its interactions with the remaining ones, hence $S_{T_i} \ge S_i$. Since Q_{peak} is a maximum, the 74.4% of the sampled parameter sets that are locally subcritical leave it at the initial value $I_H(0)$; the indices therefore characterize the sampled box and not the height of an established endemic wave.

Table 4
Weighted mean squared error (wMSE) of the climate-driven constant-coefficient model versus the free weekly model, per year.

Year	Climate-driven	Weekly free	Reduction factor
2017	85,036	618	137.6×
2018	64,959	2,553	25.4×
2019	41,436	561	73.9×
2020	23,541	359	65.6×
2021	39,449	1,552	25.4×
2022	25,046	1,711	14.6×
2023	18,360	7,598	2.4×

4.5. The constant-coefficient climate model underfits the observed dynamics

To assess how much of the seasonal variation in transmission is captured by the mechanistic, climate-driven formulation, we freed the effective transmission coefficients, allowing them to vary on a weekly basis (Section 2.10), and compared the resulting weekly model with the constant-coefficient climate-driven model. Releasing the coefficients reduced the weighted mean squared error in every year (Table 4), most notably in 2017, where the wMSE fell from 8.5×10^4 to 6.2×10^2 (a ∼137-fold reduction).

The residual between the free weekly $\beta_{h,w}, \beta_{m,w}$ and the climate-driven $a(T)b_1$, $a(T)b_2$ indicates periods in which the temperature-driven formulation fails to capture the observed transmission structure. This motivates extending the per-bite transmission probabilities to the wider set of environmental drivers introduced in Section 1.

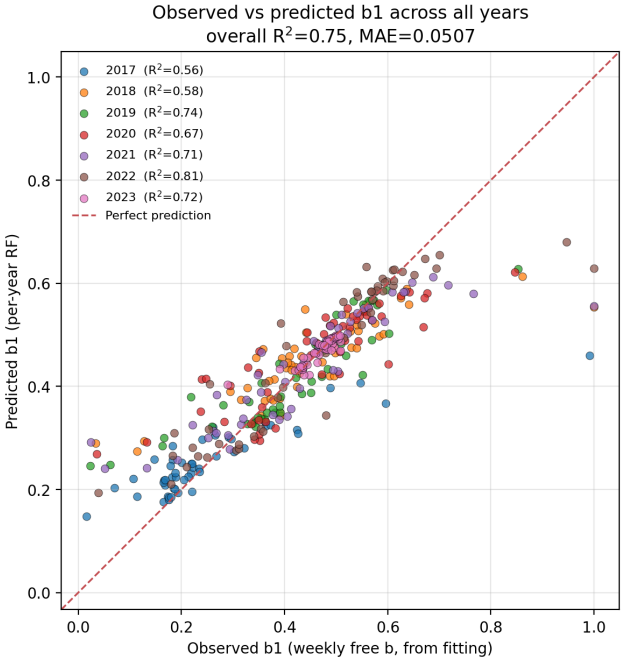
4.6. Environmental regression of the transmission probabilities b_1 and b_2

Following the framework described in Section 1, the freed weekly per-bite transmission probabilities were regressed against the environmental predictors T, R, H, D_f, F_c, F_t, representing temperature, precipitation, humidity, deforestation, fire counts and fire trends. A separate random forest model was fitted for each year.

Within each year the model explains a large part of the variance of both transmission probabilities, with prediction bias close to zero in all years ($|\text{bias}| \lesssim 0.003$ in every year; Table 5).

Table 5
Year-by-year within-year random forest performance for b_1 and b_2.

	Year	R^2	MAE	Bias	Top Driver
b_1	2017	0.563	0.0594	+0.0021	T
	2018	0.580	0.0651	−0.0005	T
	2019	0.739	0.0475	+0.0011	F_c
	2020	0.671	0.0626	+0.0021	R
	2021	0.705	0.0603	−0.0009	F_t
	2022	0.810	0.0503	+0.0010	T
	2023	0.724	0.0107	+0.0006	F_t
b_2	2017	0.531	0.0182	+0.0011	R
	2018	0.825	0.0103	+0.0005	T
	2019	0.712	0.0120	−0.0002	H
	2020	0.915	0.0105	−0.0006	F_t
	2021	0.712	0.0122	+0.0001	R
	2022	0.905	0.0103	−0.0002	F_t
	2023	0.889	0.0021	+0.0001	F_c



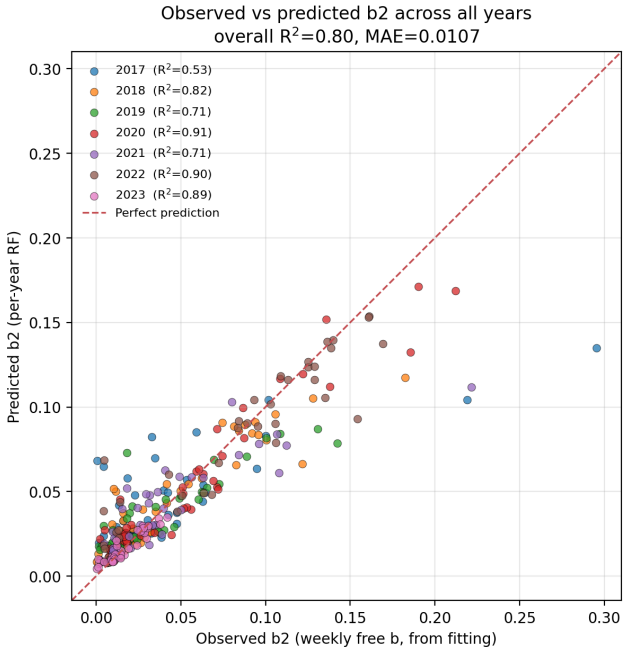


Figure 6: Scatter plot of the estimated vs predicted weekly-free proportion bites from infected humans to susceptible mosquitoes (b_1) and from infected mosquitoes to susceptible humans (b_2) that result in infection.

Table 6
Relative importance of environmental predictors for b_1 and b_2.

	b_1	b_2
Temperature (T)	20.1%	16.9%
Precipitation (R)	20.3%	16.3%
Humidity (H)	19.4%	17.1%
Deforestation (D_f)	7.4%	11.8%
Fire Counts (F_c)	14.9%	21.3%
Fire Trends (F_t)	17.9%	16.6%

The importance of the predictors, however, differs between the two probabilities, as can be seen in Table 6. Based on this, it can be seen that deforestation is consistently the weakest predictor for both b_1 and b_2, indicating that its effect is not captured at the intra-annual timescale resolved here. More strikingly, precipitation and fire counts occupy opposite ends of the two rankings: precipitation is the leading predictor of b_1 but only second-lowest for b_2, whereas fire counts lead for b_2 but rank second-lowest for b_1. Temperature and humidity, by contrast, consistently hold the second and third positions for both targets, merely exchanging order: temperature then humidity for b_1, humidity then temperature for b_2.

A leave-one-year-out cross-validation of the same forests qualifies this result. Out-of-sample performance is markedly lower, with overall $R^2 \approx 0.17$ for b_1 and ≈ 0.19 for b_2, in contrast to the strong within-year fits. This contrast confirms that the environmental drivers are strongly year-specific: the within-year seasonal trajectory of T, R, H, D_f, F_c, F_t shapes each year's transmission signal, whereas a fixed, year-independent link between the environment and b_1, b_2 is not supported by the data.

The shape of these responses is illustrated in Fig. 7, which shows the partial dependence of b_1 and b_2 on each predictor.

5. Discussion

5.1. Reproduction of the original Parham & Michael model as a negative baseline

As a diagnostic for the methodological choices adopted below, the original SIR-SEI formulation of Parham and Michael (2010) [11] was implemented with all parameter values reported in the original study, while population sizes and initial epidemiological conditions were estimated from the available rural Manaus data. Under this parameterization the infected human population declines monotonically and the mosquito population rapidly collapses to extinction, indicating that mosquito recruitment is insufficient to replace mortality. Because transmission depends on the persistence of infectious vectors, the collapse of the vector population eliminates infection in both host and vector populations. Although the delay formulation preserved positivity, it fails to reproduce the endemic behavior expected for malaria in the Amazon, confirming the need to recalibrate the environmental and transmission parameters as described in Section 2.6 and Section 2.9. This can be seen in the Supplementary Material.

5.2. The model sustains endemic transmission where the baseline does not

The transition from the delayed SIR-SEI model of Parham and Michael to the ODE-based SEIRS-SEI model was not simply a numerical convenience. As shown in Section 2.3, the original parameterization leads to the extinction of both infected humans and mosquitoes, since the linear mosquito recruitment term is not sufficient to compensate for mosquito mortality under the environmental conditions of Manaus. In the modified model, the explicit extrinsic incubation delay is replaced by the transition rate $b_{3M}\ell$, while mosquito recruitment is made density-dependent. These changes allow the model to sustain the mosquito population and reproduce endemic transmission, as established in Proposition 1. Thus, the modifications change the qualitative behavior of the system and are important for obtaining a model that is consistent with the observed persistence of malaria transmission in Manaus.

5.3. Two-factor $\mathcal{R}_0$: the transmission probabilities drive the invasion threshold

The next-generation analysis (Section 3.2) yields a reproduction number that factorises into a human-to-mosquito and a mosquito-to-human component,

$$\mathcal{R}_0 = \sqrt{\mathcal{R}_0^H \mathcal{R}_0^M},$$

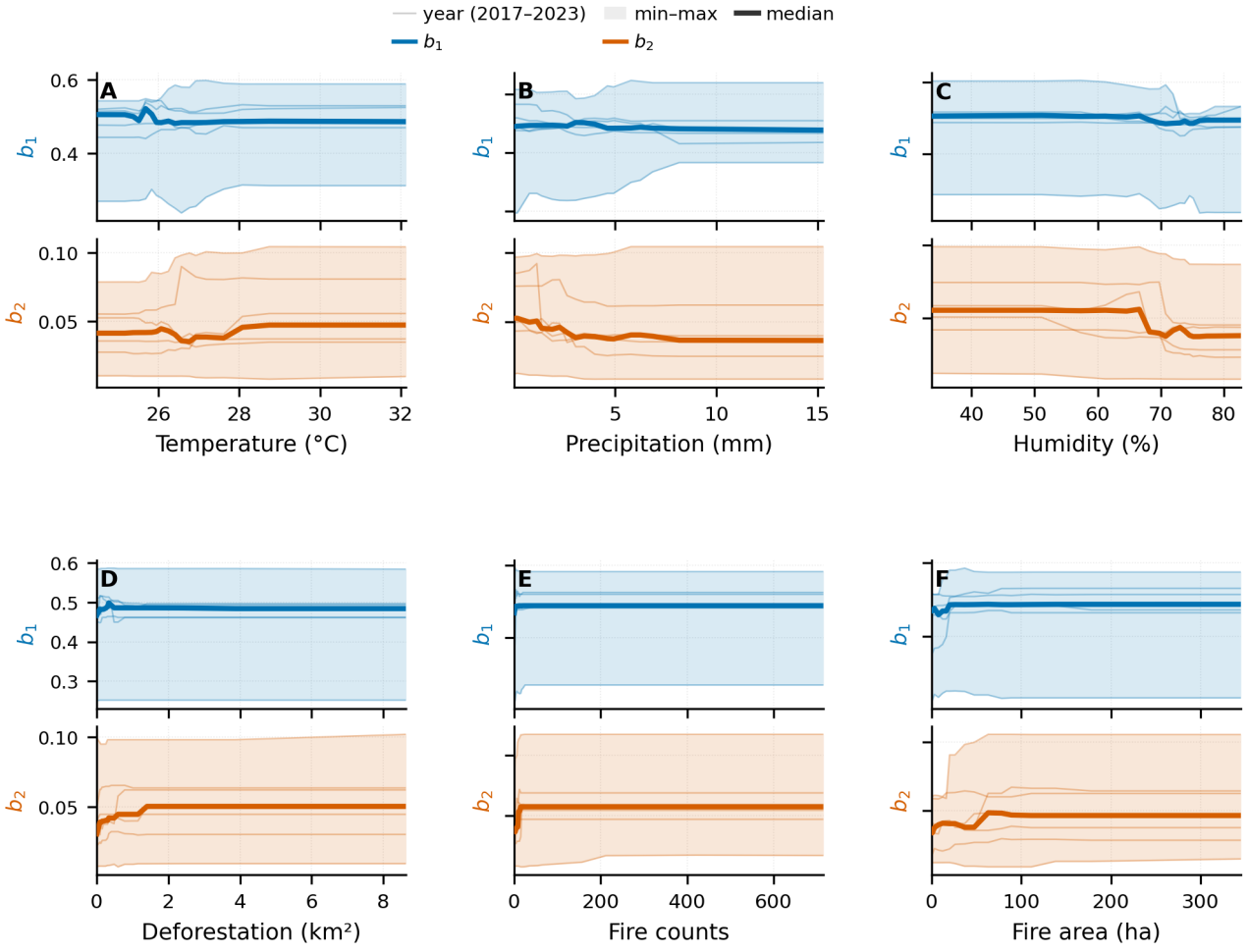


Figure 7: Partial dependence of the weekly free per-bite transmission probabilities b_1 and b_2 on each environmental predictor. Within each panel the upper and lower sub-panels share the x-axis and are drawn on their own y-scale. Each thin curve is one of the seven separately fitted per-year models (2017-2023), the thick curve is the across-year median and the shaded band spans the minimum-to-maximum range across the seven years. The x-axis covers the pooled observed range of each predictor over 2017-2023, so the curves are not extrapolated beyond the data; they are within-year partial dependences and not a fitted functional form.

a form that makes the two-step transmission cycle explicit and facilitates interpretation. The local sensitivity analysis (Section 3.3) qualifies the analytical result in an informative way: the invasion threshold is controlled by the temperature coefficients of adult mosquito survival and by the biting rate, the two transmission probabilities enter only through the square root and therefore with a universal index of $1/2$, and the intrinsic incubation period τ_H and the immunity waning rate ω are identically irrelevant to it, exactly as the factorisation predicts. In the absence of disease-induced mortality, every exposed individual eventually becomes infectious, so the latent period only shifts the timing of secondary infections, while no individual is immune at the disease-free equilibrium, so waning does not enter the linearised invasion dynamics.

Because Λ^* and M^* inherit the biting rate and carrying capacity as functions of precipitation, temperature and habitat, the reproduction number is itself a function of the environmental state. The seasonal $\mathcal{R}_0$ curve of Figure 3 makes this dependence explicit for 2017: it exceeds unity during the wet season and falls below the epidemic threshold in the driest months, confirming that the invasion risk is not constant throughout the year.

5.4. Limitations of the local sensitivity analysis

The normalized index of Eq. (38) is a local quantity: it reports the response of the output at a single reference

state and says nothing about the behaviour of the model elsewhere. Four consequences bound the conclusions of Section 4.3.

First, the index of a parameter combines two independent things, its elasticity in the output and the accuracy with which it is known. A large index attached to a well-determined parameter is reassuring; a large index attached to a weakly determined one is a warning. The decomposition therefore sharpens the identifiability problem of Section 5.8 rather than resolving it: the two most influential quantities for the seasonal peak, the coefficients A and B of the adult survival curve, are never fitted and are held at their literature values, whereas b_1 reaches its upper bound in 2020-2022 while b_2 declines monotonically to its lower bound in 2023, the two trading off against each other across years (Pearson $r = -0.83$, $p = 0.02$) as the optimizer absorbs what it cannot explain. Propagating that year-to-year spread into the closed form places $\mathcal{R}_0$ at 0.96 ± 0.19, marginally below the threshold, against 2.20 for the pre-calibration values. It is thus the fitted transmission probabilities, and not the vector vital rates, that hold the calibrated model near the edge of endemicity, and the same quantities are those that the weekly free-β diagnostic of Section 4.5 cannot reproduce from climate alone, with residual reductions of two orders of magnitude. The remaining seasonal structure of transmission is concentrated exactly where the data are least able to constrain the parameters, which is the strongest argument we can make, within this model, for treating b_1 and b_2 as time-varying quantities, as done in Section 4.6.

Second, the reference state matters. The indices are evaluated at the mean climate of the transmission season, where p_H is close to saturation and the thermal-suitability factor f_T is clamped to unity. Away from that operating point the indices of H_0, k and of the three *Anopheles* suitability temperatures change substantially, so the ranking of Table 3 describes the operating point of the calibrated model and not the whole admissible climate range; the global decomposition of Section 4.4, which spans that whole range, does reorder the parameters. The same holds for the signs: being logarithmic derivatives of a nonlinear response, the indices are informative about direction only at the point where they are evaluated.

Third, the absolute value of $\mathcal{R}_0$ quoted in Section 3.2 uses the pre-calibration carrying-capacity scale $M' = 55.99\,N = 468{,}580$, roughly nine times the value the calibration returns ($M' \approx 50{,}000$). Since M' enters the closed form only linearly, through K, its index is $1/2$ at any level and the ranking is unaffected; the discrepancy nonetheless means that the reference value of $\mathcal{R}_0$ should be read as the pre-calibration operating point and not as the calibrated one.

Finally, the two outputs answer different questions and must not be conflated. $\mathcal{R}_0$ is a disease-free quantity evaluated at a single climate-habitat state, so it measures the conditions of *invasion*, whereas the seasonal maximum measures how much malaria is actually transmitted. Within the calibrated model the parameters that decide whether $\mathcal{R}_0 > 1$ are not those that decide the height of the wave, and ranking interventions from $\mathcal{R}_0$ alone would mislead. The two immune parameters, τ_H and ω, are the one case where the algebraic and the numerical statements agree in both outputs: they are absent from Eq. (37) by construction, and their estimated contribution to the seasonal peak is below 0.05, acting on the timing of the wave rather than on its height. Within this model, refining the incubation period or the waning rate of *P. vivax* immunity would therefore not improve transmission forecasts, whereas refining the two transmission probabilities and the vector survival schedule would.

5.5. Constant coefficients underfit: the transmission probabilities are seasonally heterogeneous

The constant-coefficient model consistently underestimates the observed dynamics, as the reduction in the weighted mean squared error between the free-weekly and constant formulations demonstrates (Table 4). The improvement ranges from a ∼137-fold reduction in 2017 to a ∼2.4-fold reduction in 2023, indicating that the constant-coefficient formulation already captures most of the signal in the latter year, while the transmission probabilities vary more strongly across weeks in the former. This heterogeneity is the empirical signature of the environmental dependence of transmission, and it motivates the week-specific regression of $b_{1,w}$ and $b_{2,w}$ in Section 4.6.

5.6. Random forest: strong within years, weak in-between

The random forest regression achieves a good within-season fit (per-year R^2 ranging from 0.56 to 0.81 for b_1 and from 0.53 to 0.92 for b_2, with across-year means of 0.69 and 0.78, respectively) but a markedly poorer between-season fit (leave-one-year-out aggregate $R^2 = 0.17$ for b_1 and 0.19 for b_2). This asymmetry is informative rather than a failure of the model. Within a single year, the drivers that vary on the weekly scale (precipitation, temperature, humidity, and fire) largely determine the week-to-week fluctuations of the transmission probabilities. Across years, however, the differences are not captured by these same variables, suggesting that inter-annual variation is governed either by factors that are constant within a year in our dataset or by an underlying year-specific environmental state that is not fully represented by the contemporaneous drivers. The poor performance points to a structural limitation of the driver set rather than to an insufficiency of the statistical method.

5.7. Deforestation is a weak within-season predictor

Contrary to the expectation that land-use change drives transmission, deforestation ranked as one of the weakest predictors at the within-season scale. This does not support a

negligible role for deforestation in general, suggesting in fact that within a single year, the deforestation signal is nearly constant and therefore carries little information about week-to-week variation in the transmission probabilities.

At the between-season scale, where land-use change could plausibly matter, the random forest performs poorly, precisely because the within-year variability of deforestation does not discriminate among years. This is consistent with the hypothesis that habitat modification contributes to long-term changes in transmission intensity, even if it is not captured by a short within-season regression. The fire term, by contrast, contributes meaningfully to the variation of the mosquito-to-human probability b_2, consistent with the seasonal burning cycle in the region.

5.8. Calibration reaches the bounds

The calibrated transmission probabilities b_1 and b_2 reach the bounds of their respective ranges (Table 1), with b_1 reaching its upper bound in 2020-2022 while b_2 reaches its lower limit over the same period. This suggests that the environmental conditions affecting transmission in Manaus may not be fully captured by the original literature-based parameter ranges. It also indicates that the density-dependent mosquito recruitment and the recalibrated biting rate play an important role in reproducing the observed dynamics. At the same time, the parameters reaching their bounds highlight a limitation of the fitting procedure, suggesting that a wider parameter range or a different functional form may be needed to better capture the environmental effects on transmission.

5.9. Limitations

Several limitations should be acknowledged. First, as mentioned in Section 2.2, the ratio of rural to total population had to be recalculated, given that the ratios from the Trajetorias Project [12] for the years of 2006 and 2017 both used the estimated rural and urban populations from the 2010 census, which conceal the decline of the rural population from 2000 to 2010. In addition to that, other inconsistencies were identified in the dataset: in particular, incidence rates for the period 2015-2019 had been calculated using population estimates corresponding to 2004-2008, resulting in a temporal mismatch. These values were recalculated using population estimates consistent with the appropriate time interval. Second, the environmental drivers are treated as deterministic and observed; unobserved covariates, such as vector control interventions or local mobility, are not represented and may contribute to the between-season degradation of the random forest. Third, the model is fitted to reported case counts, which may be subject to reporting bias and under-reporting, particularly in the lowest-transmission months.

6. Conclusion

This work set out to determine whether the malaria transmission dynamics observed in rural Manaus over 2017-2023 can be explained by a climate- and environment-driven host-vector model. Starting from the delayed SIR-SEI formulation of Parham and Michael (2010), we first showed that the original parameterization cannot reproduce the observed persistence of transmission: with the reported values, mosquito recruitment is insufficient to offset mortality, the vector population collapses, and infection is extinguished in both populations (Section 5.1). The reformulation as an ODE-based SEIRS-SEI model, in which the extrinsic incubation delay is replaced by the sporogonic survival term $b_{3M}\ell$ and recruitment is made density-dependent, is therefore not a numerical convenience but a qualitative change: it restores vector persistence and allows the system to sustain the endemic transmission observed for *Plasmodium vivax* and *Anopheles darlingi* (Proposition 1). Fitted sequentially year by year, the resulting model reproduces the active-case series and its pronounced inter- and intra-annual variability (Section 4.2).

The mathematical analysis clarifies how transmission is controlled. The reproduction number factorises into a human and a vector component, $\mathcal{R}_0 = \sqrt{\mathcal{R}_0^H \mathcal{R}_0^M}$ (Eq. (37)), and depends explicitly on the environmental state through the biting rate and the carrying capacity, so that the invasion risk is not constant through the year: the seasonal curve exceeds unity during the wet season and falls below the epidemic threshold in the driest months (Figure 3). The sensitivity analyses show that this threshold is governed by the temperature coefficients of adult mosquito survival and by the biting rate, that the two transmission probabilities enter only through the square root with a universal index of $1/2$, and that the human incubation period τ_H and the immunity-waning rate ω are irrelevant to it (Section 4.3). The global decomposition of the seasonal peak reorders the parameters, as the recovery and adult-survival curvature dominate and, because most of the sampled box is locally subcritical, it should be read as a description of the parameter space rather than of an established wave (Section 4.4).

Two complementary results argue that the transmission probabilities, and not the mechanistic climate responses, are the limiting ingredient of the model. Releasing b_1 and b_2 on a weekly basis reduces the weighted mean squared error in every year, by up to two orders of magnitude, indicating that a temperature-driven constant-coefficient formulation cannot reproduce the observed seasonal structure (Section 4.5). The subsequent environmental regression then recovers much of that structure *within* each year (R^2 from 0.56 to 0.81 for b_1 and from 0.53 to 0.92 for b_2, with negligible bias) but only a small part of it *between* years (leave-one-year-out aggregate $R^2 = 0.17$ for b_1 and 0.19 for b_2), with fire activity contributing meaningfully to b_2 while deforestation remains a weak within-season predictor (Section 4.6). At the same time, the year-by-year fit drives b_1 and b_2 to their prescribed bounds and the transmission probabilities trade off against each other across years, placing the calibrated $\mathcal{R}_0$ at 0.96 ± 0.19, marginally below the threshold, against 2.20 for the pre-calibration values (Section 5.8). It is thus the fitted transmission probabilities, rather than the vector

vital rates, that hold the calibrated model near the edge of endemicity.

Taken together, these findings indicate that environmental drivers act on the Amazonian transmission cycle mainly by modulating the effective per-bite transmission probabilities, and that a faithful description of the seasonal dynamics requires these coefficients to be treated as time-varying rather than constant. This has two practical implications. First, because the parameters that decide whether $\mathcal{R}_0 > 1$ are not the same as those that decide the height of the seasonal wave, intervention strategies ranked from $\mathcal{R}_0$ alone may be misleading, so refining the vector survival schedule and the transmission probabilities would improve forecasts more than refining the immune parameters. Second, the systematic drift of the fitted transmission probabilities towards their bounds underlines the need for wider, data-informed parameter ranges or alternative functional forms.

Several limitations remain. The environmental drivers are treated as deterministic and observed, so unmodelled factors such as vector-control interventions and local mobility are absorbed by the fitted parameters; the model is fitted to reported case counts, which are subject to reporting bias and under-reporting, particularly in the lowest-transmission months; and the demographic corrections described in Section 2.2 reflect data inconsistencies that propagate into the estimates of transmission intensity (Section 5.9). Future work should therefore incorporate these unobserved covariates, couple the empirical regression of b_1 and b_2 back into the mechanistic model, and assess the transferability of the calibrated coefficients to other municipalities and years of the Brazilian Amazon.

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Supplementary Material

All files of data collection and treatment, as well as results obtained throughout the development of this work, are available at `https://github.com/RaphaLevy/Environmental_Drivers_of_Malaria`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used EpidBot, developed by Kwar-AI [42], to assist the implementation and support the parameterisation entomological functions of the SEIRS-SEI model, to discuss the statistical analysis and with drafting specifications for the sensitivity-analysis code, and to review the manuscript for clarity and language; and OpenCode Zen (model `Big Pickle`) to assist with scientific-computing code and LaTeX manuscript editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix

A. Tables

Table A.1
Response functions and derived quantities of the model.

Parameter	Definition	Formulation
$p(T)$	Daily mosquito survival rate dependent on temperature	$\exp\left[-1/(AT^2+BT+C)\right]$
$p(H)$	Daily mosquito survival rate dependent on humidity [1]	$\left(\varphi\exp\left[-k(H-H_0)\right]+1\right)/\left(\exp\left[-k(H-H_0)\right]+1\right)$
$p(T,H)$	Daily mosquito survival rate	$p(T)p(H)$
$p_L(R)$	Probability of larval survival dependent on rainfall [2]	$(4p_{ML}/R_L^2)R(R_L-R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$\exp\left[-(c_1T+c_2)\right]$
$p_L(R,T)$	Probability of larval survival	$p_L(R)p_L(T)$
$\mu(T,H)$	Mosquito mortality rate per capita (days^{-1})	$-\log(p(T,H))$
$a(T)$	Biting rate (days^{-1})	$(T-T')/D_1$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD/(T-T_{\min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1/(c_1T+c_2)$
$\ell(\tau_M)(T,H)$	Probability of mosquito survival during sporozoite cycle (days^{-1})	$p(T,H)^{\tau_M(T)}$
$b(R,T)$	Mosquito birth rate (days^{-1})	$B_Ep_E(R)p_L(R,T)p_P(R)/(\tau_E+\tau_L(T)+\tau_P)$
$M(t)$	Total population of mosquitoes	$S_M(t)+E_M(t)+I_M(t)$
$N(t)$	Total population of humans	$S_H(t)+E_H(t)+I_H(t)+R_H(t)$

[1] $p(H)$ was added in order to take into consideration the deleterious effects of humidity. Unlike for T and R, which use the daily average value of temperature and precipitation, respectively, H uses the daily minimal value of humidity.

[2] $p_E(R)$ and $p_P(R)$ have an analogous expression to $p_L(R)$, substituting p_{ML} for p_{ME} and p_{MP} respectively.

Table A.2
List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection: 0.04	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection: 0.09	Dimensionless
γ	1/Average duration of infectiousness in humans: 1/120	days^{-1}
ω	1/Average duration of protective immunity after infection: 1/270	days^{-1}
B_E	Number of eggs laid per adult per oviposition: 200	Dimensionless
p_{ME}	Maximum probability of egg survival: 0.9	Dimensionless
p_{ML}	Maximum probability of larval survival: 0.25	Dimensionless
p_{MP}	Maximum probability of pupal survival: 0.75	Dimensionless
τ_E	Duration of the egg development phase: 1	days
τ_P	Duration of the pupal development phase: 1	days
τ_H	Latent period of infection within humans: 10	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 [3]	mm
T_{min}	Minimum temperature, below which there is no development of the parasite: 14.5	$°C$
T_{max}	Maximum temperature, above which there is no development of the parasite: 29.8 [35]	$°C$
T_{opt}	Optimal temperature for development of the parasite: 23.5 [35]	$°C$
$T_{\text{min A.}}/T'$	Minimum temperature, below which there is no development of the vector: 16 [4] [24, 36]	$°C$
$T_{\text{max A.}}$	Maximum temperature, above which there is no development of the vector: 34 [24, 36, 37]	$°C$
$T_{\text{op A.}}$	Optimal temperature for transmission: 25 [24]	$°C$
DD	Degree-days for parasite development. “Sum of heat" for maturation: 105 [35, 38]	$°C$ days
A	Empirical sensitivity parameter: -0.03	$(°C^2 \text{ days})^{-1}$
B	Empirical sensitivity parameter: 1.31	$(°C \text{ days})^{-1}$
C	Empirical sensitivity parameter: -4.4	days^{-1}
D_1	Temperature scaling factor of the biting rate: 24.8	$°C$ days
c_1	Empirical sensitivity parameter: 0.00554	$(°C \text{ days})^{-1}$
c_2	Empirical sensitivity parameter: -0.06737	days^{-1}
b_{3M}	Sporogonic development rate in exposed mosquitoes (Eq. 4), normalized by $b_{3M}(T_{\text{opt}}) = 1/\tau_M(T_{\text{opt}})$ [5]	days^{-1}
b_{3H}	Infection rate in exposed humans ($1/\tau_H$): 1/10	days^{-1}
H_0	Critical humidity threshold for 50% survival transition: 58	%
φ	Baseline survival probability at extreme low humidity: 0.05	Dimensionless
k	Steepness of the humidity-survival sigmoid response: 0.25	$\%^{-1}$
M'	Baseline mosquito population scale of the carrying capacity $K(t) = M' f_T(T) f_R(R)$: 468,580 [6]	Individuals

[3] The reference rainfall level R_L for the Amazon was approximated based on reported associations between monthly precipitation and malaria incidence [39]. In particular, precipitation levels in the range of 917 to 1019 mm/month were associated with a decrease in malaria cases. The midpoint of this interval (968 mm/month) was used as a representative value and converted to a daily rate by dividing by 30 days, yielding $R_L \approx 32.3$ mm/day. This value is therefore interpreted as the threshold scale for high rainfall conditions in the model.

[4] In Parham and Michael [11], this parameter is denoted by T_1. Here, we adopt a different notation to avoid confusion with the mean temperature in the absence of seasonality, also denoted by T_1 in the original paper. The biting-rate threshold T' was set equal to the minimum temperature required for vector development rather than fitted independently, so that $T' = T_{\text{min A.}} = 16\,°C$ in the baseline parameterization. The two remain distinct parameters of distinct response functions, $a(T)$ and $f_T(T)$, and are perturbed independently in Section 3.3.

[5] b_{3M} and b_{3H} are not free parameters: they are the development rates $b_{3H} = 1/\tau_H$ and b_{3M} of Eq. 4, the latter being a function of DD, T_{min} and T_{max} once η is fixed by the normalization above.

[6] M' is one of the nine parameters calibrated year by year (Section 2.9). The value shown is the reference value used for the analytic decomposition of $\mathcal{R}_0$ (Section 3.3); its admissible range is that of Table A.3.

Table A.3
Admissible parameter box of the sensitivity analyses. The 25 decomposed parameters, grouped by mechanism as in Section 3.3, with the baseline value of Table A.2 and the range over which the global decomposition of Section 4.4 samples them. Every range brackets the baseline value. The nine parameters calibrated year by year (Table 1) are sampled over the bounds used in the calibration.

Parameter	Baseline	Sampled range
Transmission probabilities and capacity scale		
b_1	0.04	[0.001, 0.5]
b_2	0.09	[0.01, 0.5]
M'	468,580	[50,000, $500\,N_y$]
Human immunity and turnover		
τ_H (days)	10	[10, 20]
γ (days^{-1})	1/120	[1/150, 1.0]
ω (days^{-1})	1/270	[1/300, 0.1]
Humidity-dependent adult survival		
H_0 (%)	58	[20, 75]
k (%$^{-1}$)	0.25	[0.01, 1.0]
φ	0.05	[0.01, 0.5]
Biting rate		
T' (°C)	16	[14, 20]
D_1 (°C days)	24.8	[18, 32]
Sporogonic development (Plasmodium)		
DD (°C days)	105	[80, 140]
$T^{\mathrm{p}}_{\mathrm{min}}$ (°C)	14.5	[12, 17]
$T^{\mathrm{p}}_{\mathrm{max}}$ (°C)	29.8	[27, 32]
$T^{\mathrm{p}}_{\mathrm{opt}}$ (°C)	23.5	[21, 26]
Immature-stage survival and recruitment		
R_L (mm)	32.3 [7]	[20, 50]
B_E	200	[100, 300]
c_1 (°C^{-1} days^{-1})	0.00554	[0.003, 0.008]
c_2 (days^{-1})	−0.06737	[−0.09, −0.04]
Adult survival curve and mortality		
A (°C^2 days^{-1})	−0.03	[−0.05, −0.01]
B (°C^{-1} days^{-1})	1.31	[1.0, 1.6]
C (days^{-1})	−4.4	[−5.5, −3.5]
Anopheles thermal suitability		
$T^{\mathrm{a}}_{\mathrm{min}}$ (°C)	16	[13, 19]
$T^{\mathrm{a}}_{\mathrm{opt}}$ (°C)	25	[22, 28]
$T^{\mathrm{a}}_{\mathrm{max}}$ (°C)	34	[30, 38]

[7] N_y denotes the estimated rural population of Manaus for year y. T' and $T^{\mathrm{a}}_{\mathrm{min}}$ share the baseline value 16 °C (note 4 of Table A.2) but are distinct parameters and are sampled from distinct ranges, so that the two response functions they enter keep an independent degree of freedom.

B. Derivation of the basic reproduction number

This appendix details the next-generation matrix (NGM) computation summarised in Section 3.2. The infected compartments are ordered as (E_H, I_H, E_M, I_M), so $m = 4$, and all parameters are as defined in Section 3.

Disease-free states and the NGM formalism: Following van den Driessche and Watmough [33], let X_s be the set of all disease-free states:

$$X_s = \{x \geq 0 | x_i = 0,\ i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^\top$, $x_i \geq 0$ is the number of individuals in compartment i, and every function is assumed to be continuously differentiable at least twice in each variable (C^2).

Rearranging the model equations so that the first m components are the infected compartments, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i, $\mathcal{V}_i^+(x)$ the rate of transfer into compartment i by all other means, and $\mathcal{V}_i^-(x)$ the rate of transfer out of compartment i. The model can then be written as

$$\dot{x}_i = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), \quad i = 1, \dots, n,$$

with $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. The next-generation matrices are the Jacobians of these terms evaluated at the disease-free equilibrium x_0:

$$F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j}\right], \quad V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j}\right], \quad 1 \leq i, j \leq m,$$

so that $\mathcal{R}_0 = \rho(FV^{-1})$.

Disease-free equilibrium and mosquito demographic equilibrium: At the DFE the human population is fully susceptible, $S_H^* = N$, and all infected compartments vanish. The mosquito population rests at the demographic equilibrium M^* of the density-dependent recruitment dynamics (Eq. 12). Because the climate drivers are time-dependent, we evaluate them at their fixed reference state (the study-period means $\bar{T}, \bar{R}$), so that the recruitment rate and the carrying capacity become constants:

$$\bar{b} = b(\bar{R}, \bar{T}), \quad \bar{K} = K(t)\Big|_{\text{DFE}} = M'\, f_T(\bar{T})\, f_R(\bar{R}),$$

and

$$\frac{dM}{dt} = \bar{b}\,\bar{K}\left(1 - \frac{M}{\bar{K}}\right) - \mu M$$

Imposing $\Lambda(M^*) = \mu M^*$ gives

$$\bar{b}\,\bar{K}\left(1 - \frac{M^*}{\bar{K}}\right) = \mu M^* \quad \Longrightarrow \quad M^* = \frac{\bar{b}\,\bar{K}}{\bar{b} + \mu},$$

and hence the recruitment at the DFE is

$$\Lambda^* = \mu M^* = \frac{\mu\,\bar{b}\,\bar{K}}{\bar{b} + \mu}$$

By Proposition 1 (Section 3.1), $M^* > 0$ is the unique relevant equilibrium whenever $\bar{b}, \bar{K} > 0$. Therefore, the disease-free equilibrium will be

$$x_0 = \left(N,\ 0,\ 0,\ 0,\ M^*,\ 0,\ 0\right)^\top. \tag{46}$$

Verification of $\mathcal{R}_0$ by explicit multiplication: The new-infection and transfer matrices, evaluated at the DFE, are

$$F = \begin{pmatrix} 0 & 0 & 0 & \dfrac{ab_2 S_H^*}{N} \\ 0 & 0 & 0 & 0 \\ 0 & \dfrac{ab_1 S_M^*}{N} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{47}$$

$$V = \begin{pmatrix} b_{3H} & 0 & 0 & 0 \\ -b_{3H} & \gamma & 0 & 0 \\ 0 & 0 & b_{3M}\ell + \mu & 0 \\ 0 & 0 & -b_{3M}\ell & \mu \end{pmatrix},$$

where $S_H^* = N$ and $S_M^* = M^*$ at the DFE.

From (47), the transfer matrix is block-diagonal, $V = \operatorname{diag}(V_H, V_M)$, with

$$V_H = \begin{pmatrix} b_{3H} & 0 \\ -b_{3H} & \gamma \end{pmatrix}, \qquad V_M = \begin{pmatrix} b_{3M}\ell + \mu & 0 \\ -b_{3M}\ell & \mu \end{pmatrix},$$

so that

$$V^{-1} = \operatorname{diag}\left(V_H^{-1}, V_M^{-1}\right),$$

$$V_H^{-1} = \begin{pmatrix} \frac{1}{b_{3H}} & 0 \\ \frac{1}{\gamma} & \frac{1}{\gamma} \end{pmatrix},$$

$$V_M^{-1} = \begin{pmatrix} \frac{1}{b_{3M}\ell + \mu} & 0 \\ \frac{b_{3M}\ell}{\mu(b_{3M}\ell + \mu)} & \frac{1}{\mu} \end{pmatrix}$$

Carrying out the block multiplication FV^{-1} yields the hollow form

$$FV^{-1} = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix},$$

$$P = \begin{pmatrix} \frac{ab_2 S_H^* b_{3M}\ell}{N\mu(b_{3M}\ell + \mu)} & \frac{ab_2 S_H^*}{N\mu} \\ 0 & 0 \end{pmatrix},$$

$$Q = \begin{pmatrix} \frac{ab_1 S_M^*}{N\gamma} & \frac{ab_1 S_M^*}{N\gamma} \\ 0 & 0 \end{pmatrix}$$

Both P and Q have rank one, so PQ has a single nonzero eigenvalue $\lambda = \frac{a^2 b_1 b_2 S_H^* S_M^* b_{3M}\ell}{N^2\mu\gamma(b_{3M}\ell + \mu)}$, and the spectrum of FV^{-1} is

$$\sigma\left(FV^{-1}\right) = \left\{0,\ 0,\ \pm\sqrt{\lambda}\right\} = \left\{0,\ 0,\ \pm\sqrt{\mathcal{R}_0^H\,\mathcal{R}_0^M}\right\}$$

Substituting $S_H^* = N$ and $S_M^* = M^* = \Lambda^*/\mu$ into λ and taking the square root reproduces (36).

Spectral computation of $\rho(FV^{-1})$*:* The matrix V is block-diagonal in the human-mosquito partition, with blocks V_H and V_M, so

$$V^{-1} = \begin{pmatrix} \frac{1}{b_{3H}} & 0 & 0 & 0 \\ \frac{1}{\gamma} & \frac{1}{\gamma} & 0 & 0 \\ 0 & 0 & \frac{1}{\mu + b_{3M}\ell} & 0 \\ 0 & 0 & \frac{b_{3M}\ell}{(\mu + b_{3M}\ell)\,\mu} & \frac{1}{\mu} \end{pmatrix}$$

Multiplying F by V^{-1} we obtain

$$FV^{-1} = \begin{pmatrix} 0 & 0 & \frac{ab_2 b_{3M}\ell}{(\mu + b_{3M}\ell)\mu} & \frac{ab_2}{\mu} \\ 0 & 0 & 0 & 0 \\ \frac{ab_1 M^*}{N\gamma} & \frac{ab_1 M^*}{N\gamma} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The second and fourth rows are zero, so the spectral problem reduces to the 2×2 block coupling the human and mosquito exposed classes,

$$\left[FV^{-1}\right]_{\{1,3\}} = \begin{pmatrix} 0 & \frac{ab_2 b_{3M}\ell}{(\mu + b_{3M}\ell)\mu} \\ \frac{ab_1 M^*}{N\gamma} & 0 \end{pmatrix},$$

whose characteristic equation is

$$\lambda^2 = \frac{a^2 b_1 b_2 b_{3M}\ell\, M^*}{N\gamma\,(\mu + b_{3M}\ell)\,\mu}$$

The eigenvalues of FV^{-1} are $\lambda \in \left\{ 0,\ 0,\ \pm\sqrt{\mathcal{R}_H \mathcal{R}_M} \right\}$, where

$$\mathcal{R}_H = \frac{ab_2}{\gamma}, \quad \mathcal{R}_M = \frac{ab_1 \Lambda^*}{N\mu} \frac{b_{3M}\ell}{(b_{3M}\ell + \mu)\,\mu},$$

so that $\mathcal{R}_0 = \rho(FV^{-1}) = \sqrt{\mathcal{R}_H \mathcal{R}_M}$. Finally, substituting $\Lambda^* = \mu M^*$ recovers the expression in Section 3.2:

$$\mathcal{R}_0 = \sqrt{\frac{a^2 b_1 b_2\, \Lambda^*\, b_{3M}\ell}{N\gamma\,(b_{3M}\ell + \mu)\,\mu^2}} \tag{48}$$